

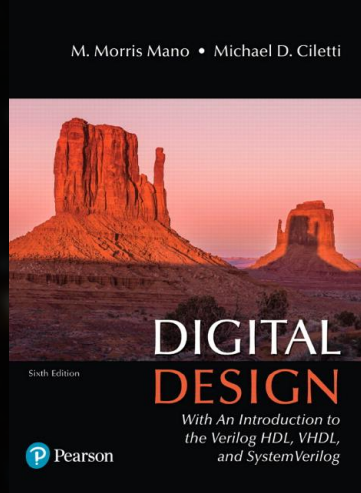
RECORDING



LEC02 & LAB02

Week02 >> Lec02: Number Systems

Week02 >> Lab02: Programming Environment Setup

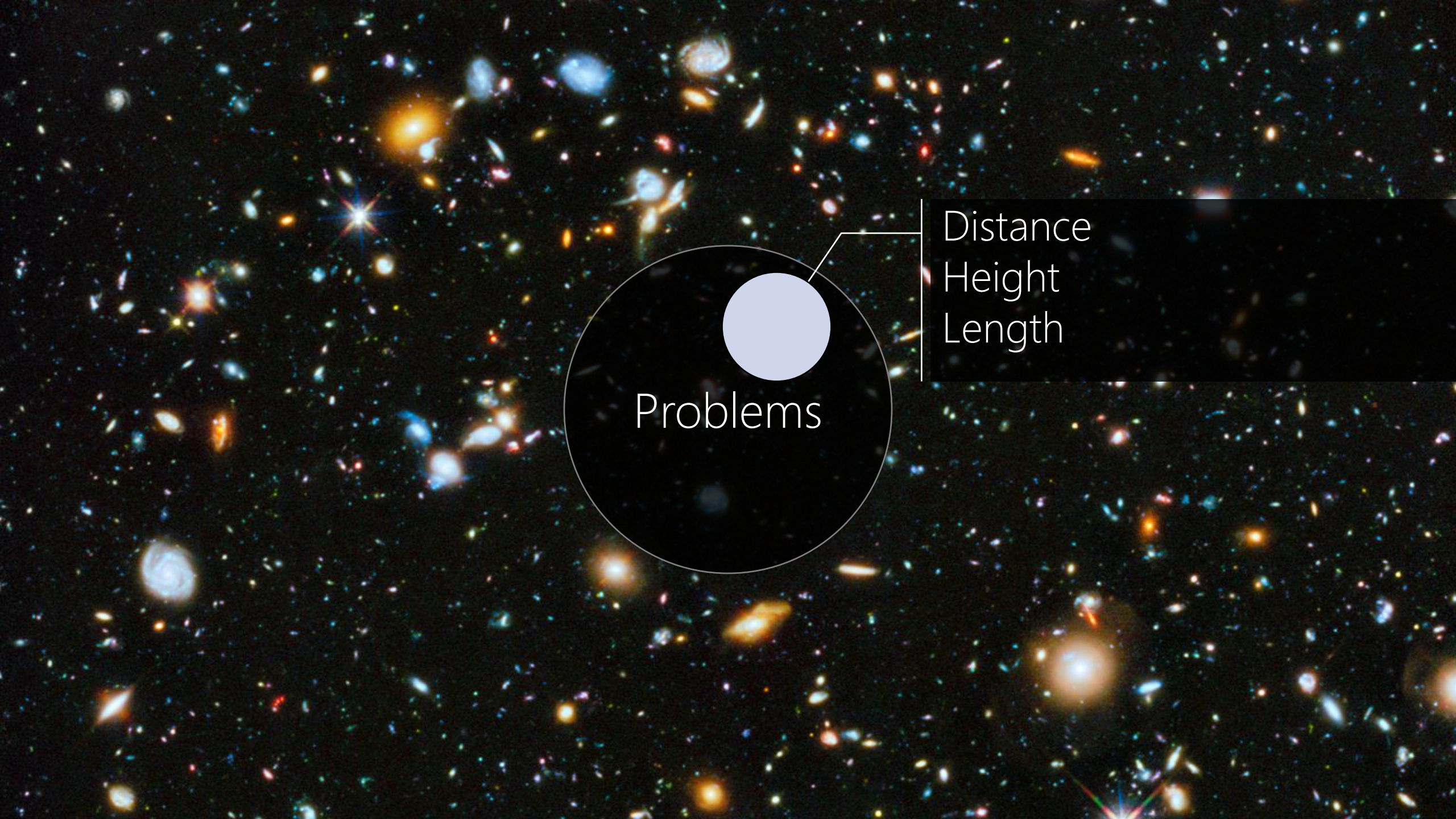


Chapter 1

Digital Systems and Binary Numbers

The background is a deep-field astronomical image, likely from the Hubble Space Telescope, showing a vast number of galaxies in various colors (blue, orange, white) against a black sky. The galaxies are of various shapes and sizes, some appearing as bright, distinct objects and others as faint, distant specks. A large, solid black circle is centered in the image. Inside this circle, the word "Problems" is written in a white, sans-serif font. Below the word, there is a red, wavy line that resembles a handwritten underline.

Problems

The background is a deep space photograph showing a dense field of galaxies in various colors (blue, orange, white) against a black sky. A diagram is overlaid on the right side. It consists of a large black circle with a smaller light blue circle inside it. The word "Problems" is written in white inside the large circle. A line extends from the top of the light blue circle to a black rectangular box on the right. Inside this box, the words "Distance", "Height", and "Length" are listed vertically in white.

Problems

Distance
Height
Length



Quantization



I promise I will be so incredibly careful.
I will be incredibly careful with it. I promise.

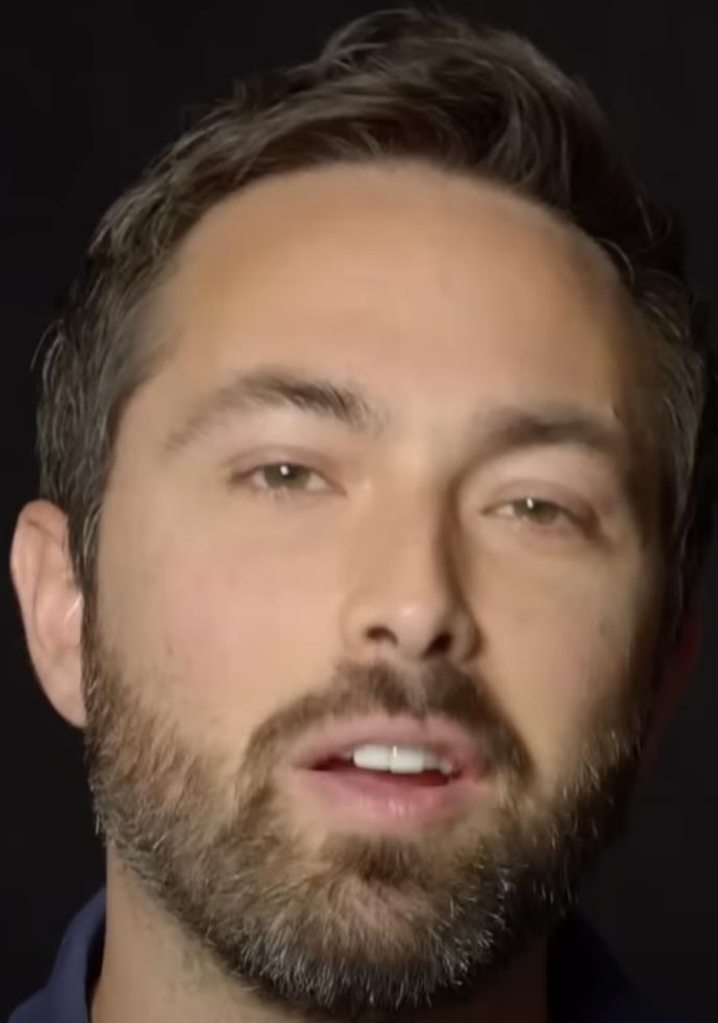


0:12 / 11:43



HD





The kilogram is in trouble.



0:21 / 9:48 • The kilogram >



HD





DISCRETE SYSTEMS

UNARY SYSTEM

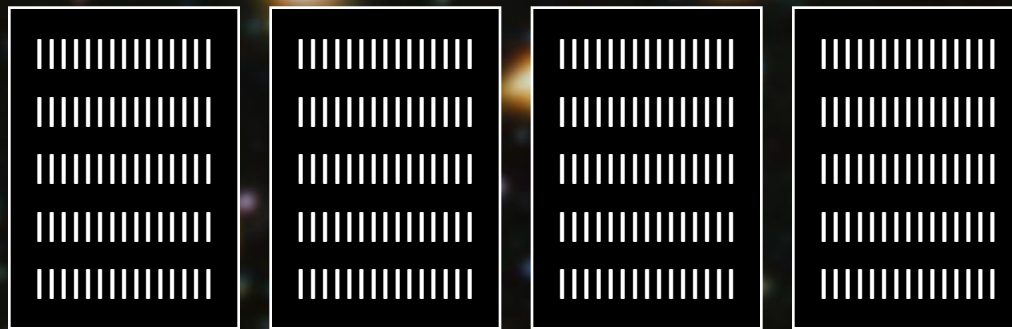
aka. Base-1

I II III IV V VI VII VIII IX X

Roman Numerals
Originated in Ancient Rome
8th Century BC

$\sim 150 \text{ million km} \div \sim 13,000 \text{ km} = \sim 12,000 \text{ Earth}$
paper = $\sim 3,000$ positions
 $12,000 \div 3,000 = 4 \text{ pages!}$

Poor Data
Model



Simple/Fast
Algorithm

A deep-field astronomical image showing a vast field of galaxies and stars against a black background. The galaxies are in various colors, including blue, orange, and white, and are scattered across the frame. Two horizontal blue lines are positioned above and below the text.

NUMBER SYSTEMS

— = ≡ ≠ ୧ | ୬ ୭ ୫ ୨

Brahmi

3rd and 7th century AD

୨ ୩ ୪ ୫ ୬ | ୭ ୮ ୯ ୦

Hindu (Gwalior)

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

Sanskrit-Devanagari

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯

Western Arabic (Gobar)

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

Eastern Arabic

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

11th Century (Apices)

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

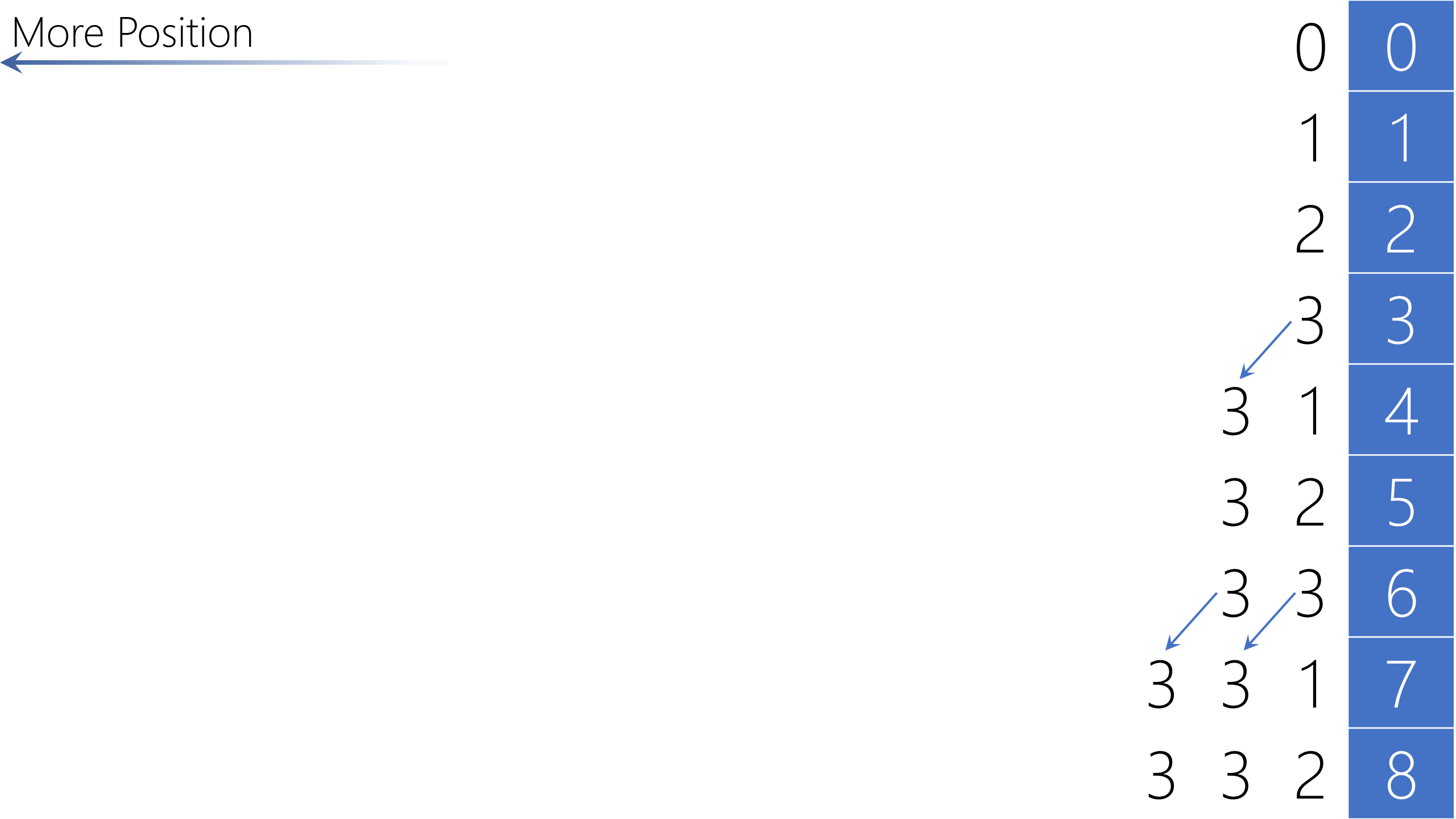
15th Century

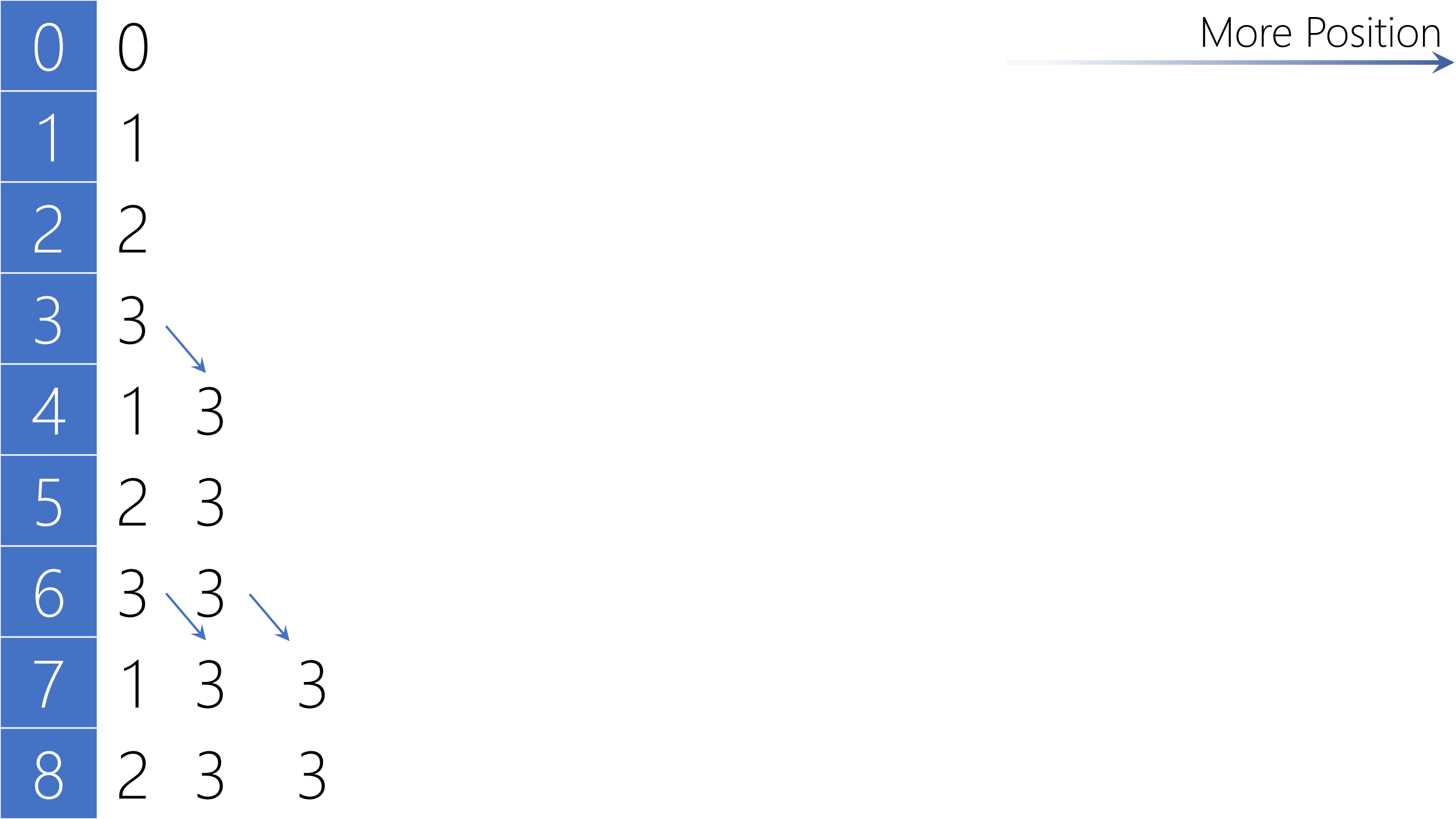
୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

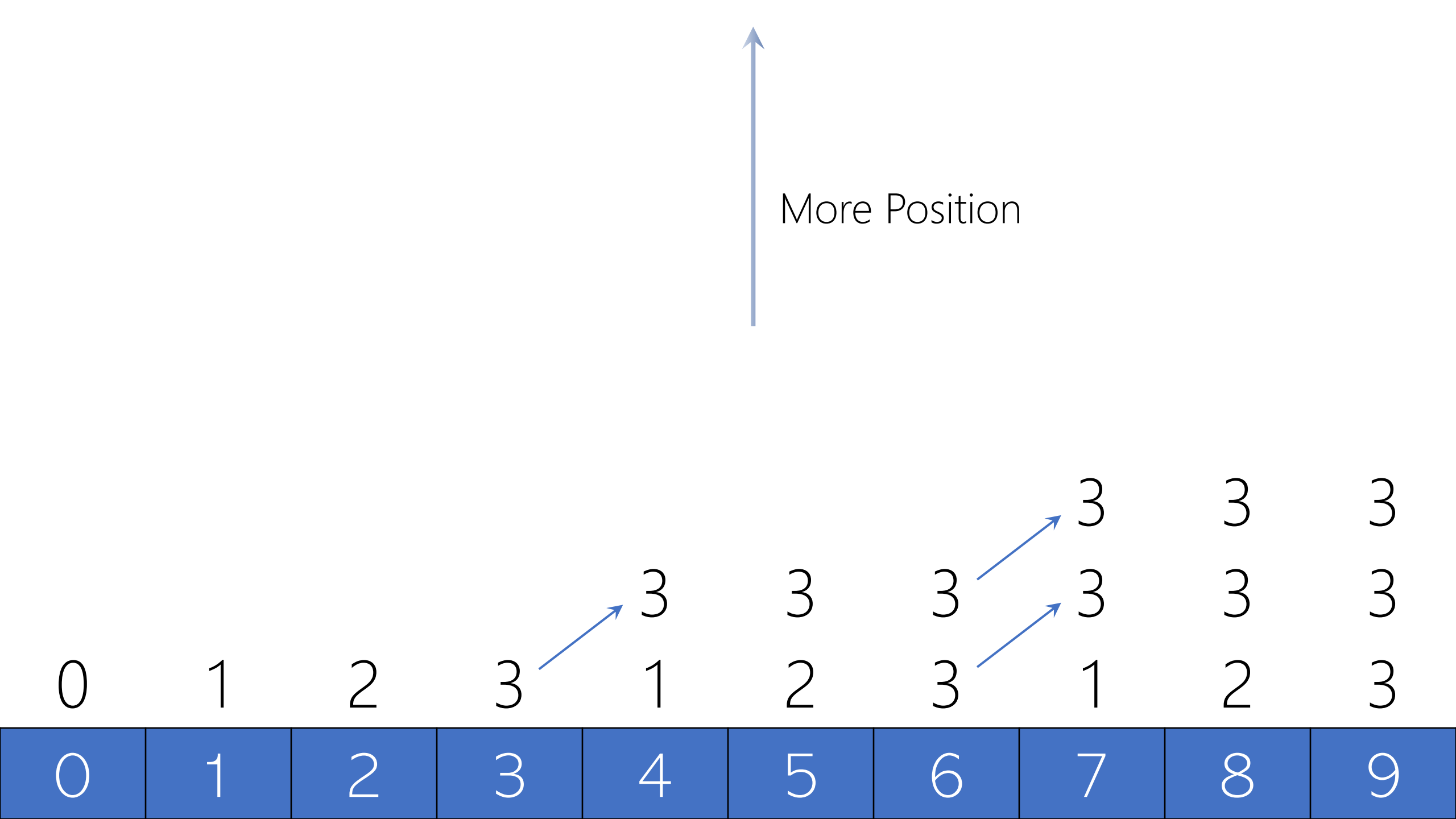
16th Century (Dürer)

0123456789

Hossein's Number System





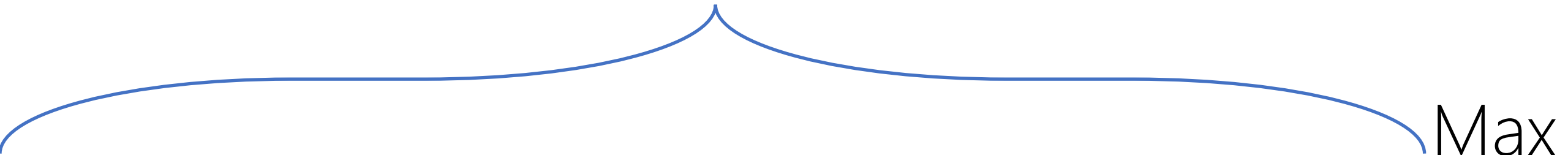


More Position			0	0=0
			1	1=1
			2	2=2
			3	3=3
			3 1	3+1=4
			3 2	3+2=5
			3 3	3+3=6
			3 3 1	3+3+1=7
			3 3 2	3+3+2=8

0	0	...			0	0	0
---	---	-----	--	--	---	---	---



n positions

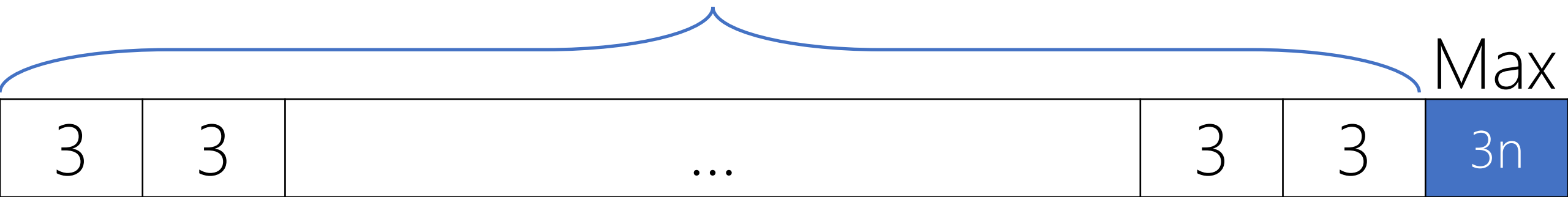


3	3	...			3	3	?
---	---	-----	--	--	---	---	---

What is the max number given n position?

$$3+3+3+\cdots+3 = 3 \times n = 3n$$

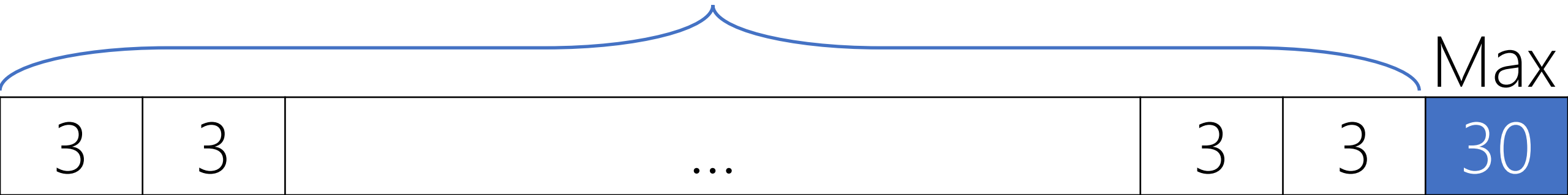
n positions



$$\text{Max} = 3n$$

$$n=10 \Rightarrow (3 \times 10) = 30$$

10 positions

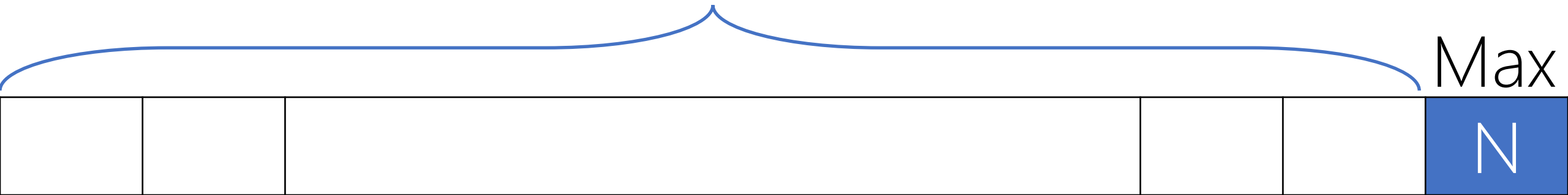


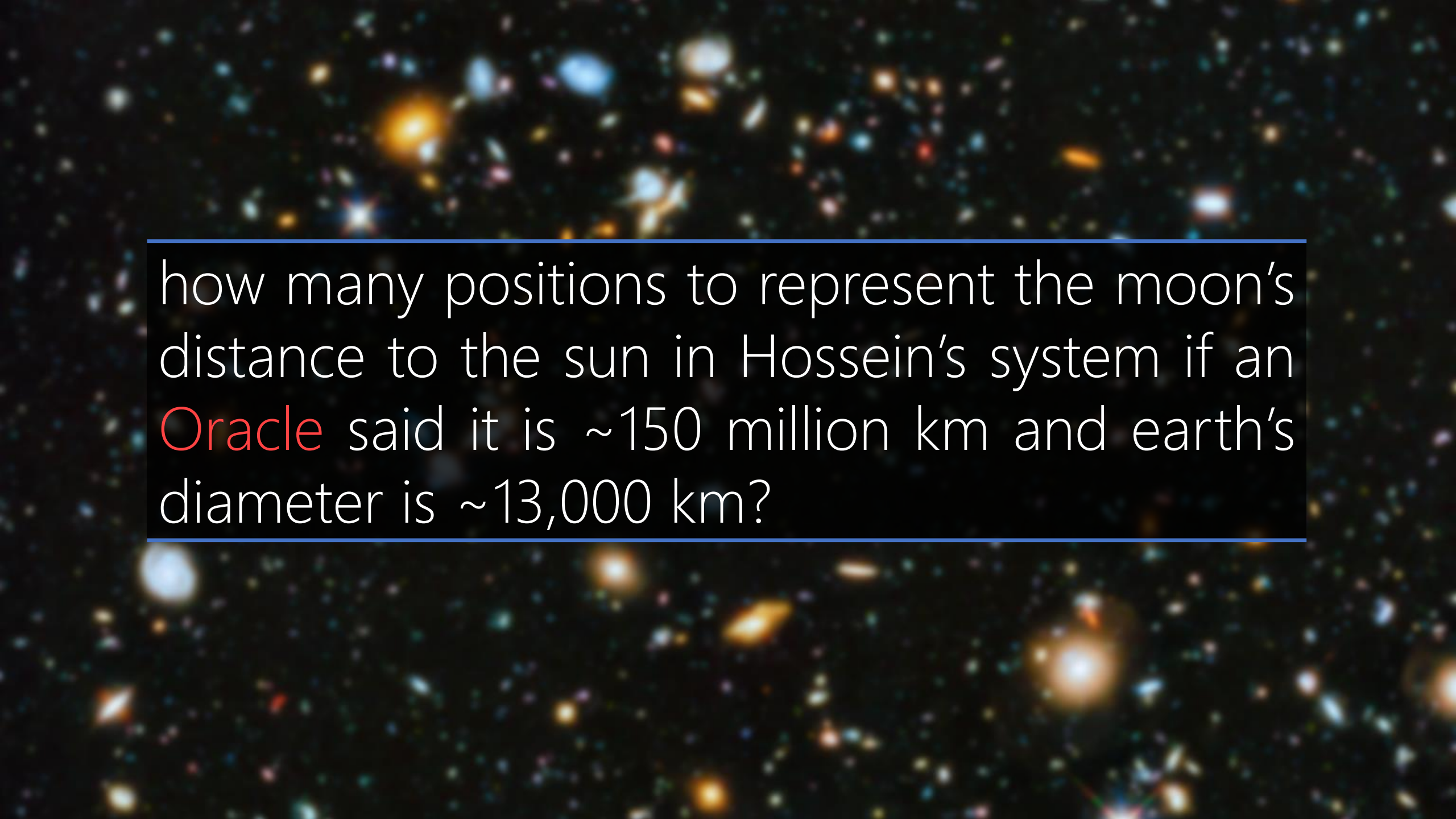
The other way around,
What is number of required position for a number N?

$$\text{Max} = N = 3n$$

$$N \div 3 = n$$

? positions





how many positions to represent the moon's distance to the sun in Hossein's system if an **Oracle** said it is ~150 million km and earth's diameter is ~13,000 km?

$\sim 150 \text{ million km} \div \sim 13,000 \text{ km} = \sim 12,000 \text{ Earth}$

$$N = 12,000$$

$$n = N \div 3 = 12,000 \div 3 = 4,000 \text{ positions}$$

paper = $\sim 3,000$ positions

$$4,000 \div 3,000 = 1.33 \text{ pages} = 2 \text{ pages}$$

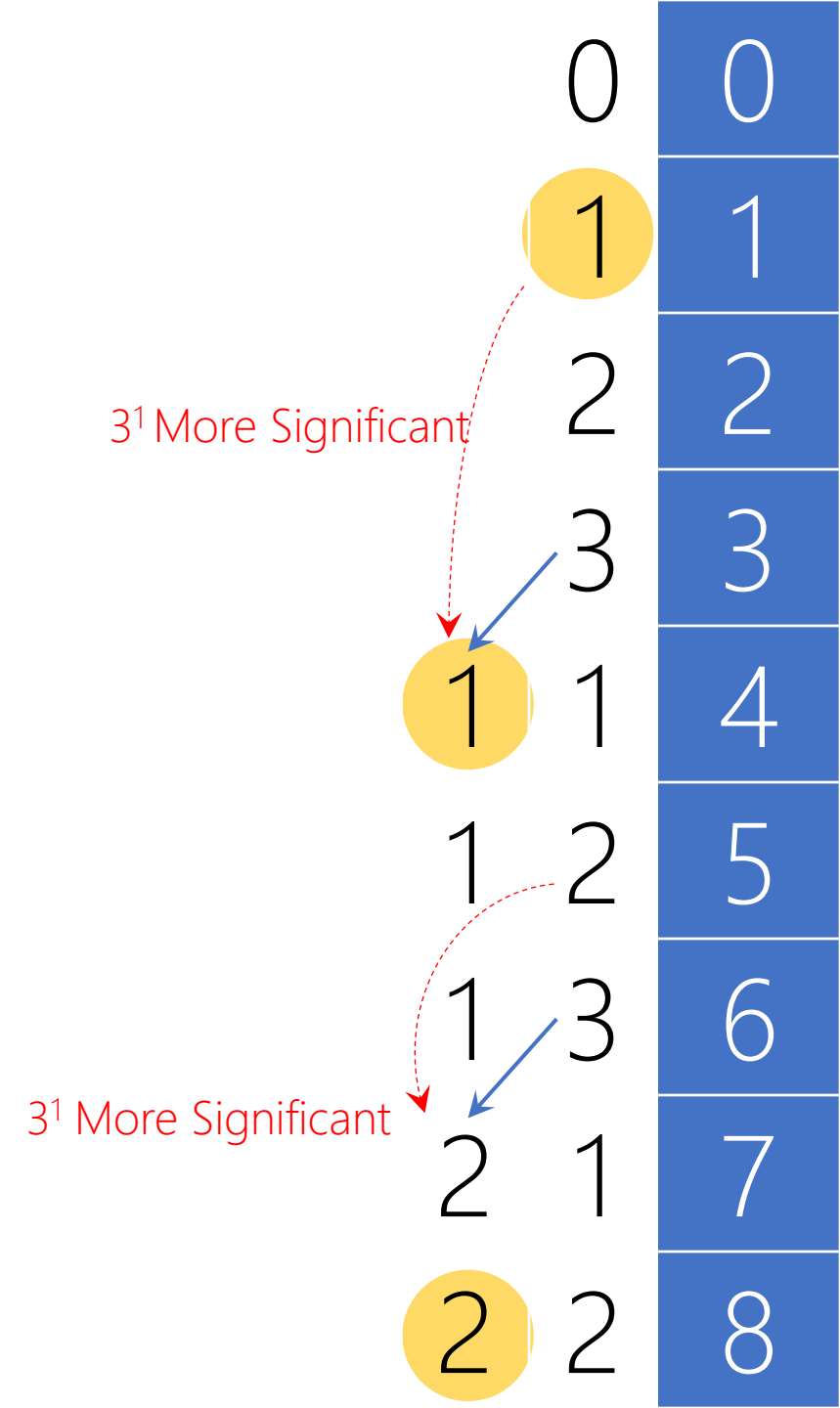


More Position				0	0
				1	1
				2	2
				3	3
1 rounds of all (3) symbols: 1 × 3		3	1	4	
		3	2	5	
		3	3	6	
2 rounds of all (3) symbols: 2 × 3	3	3	1	7	
	3	3	2	8	

More Significant Position
←

1 round of all (3) symbols = 1×3^1

2 rounds of all (3) symbols = 2×3^1



More Significant Position			2	2	8
			2	3	9
			3	1	10
			3	2	11
			3	3	12
			?	?	13
			?	?	14
			?	?	15
			?	?	16

← More Significant Position

1 rounds of all (3) symbols in the 2nd significant position
 $= 1 \times 3 \times 3^1 = 1 \times 3^2$

2	2	8
2	3	9
3	1	10
3	2	11
3	3	12
1	?	13
?	?	14
?	?	15
?	?	16

← More Significant Position

1 rounds of all (3) symbols in the 1st significant position
 $= 1 \times 3 = 1 \times 3^1$

1 rounds of all (3) symbols in the 2nd significant position
 $= 1 \times 3 \times 3^1 = 1 \times 3^2$

2	2	8
2	3	9
3	1	10
3	2	11
3	3	12
1	1	13
?	?	14
?	?	15
?	?	16

More Significant Position				2	2	8
				2	3	9
				3	1	10
				3	2	11
				3	3	12
				1	1	13
				?	?	14
				?	?	15
				?	?	16

0123456789

~~Hossein's Number System~~

Christina Reynolds-Badder (2021)

THE VERTICES NUMBER SYSTEM

0, $\wedge$, $\sqcap$, Δ , $\diamond$, $\star$

0 $\rightarrow$ 0	12 $\rightarrow$ $\sqcap$ 0
1 $\rightarrow$ $\wedge$	13 $\rightarrow$ $\sqcap$ $\wedge$
2 $\rightarrow$ $\sqcap$	14 $\rightarrow$ $\sqcap$ $\sqcap$
3 $\rightarrow$ Δ	15 $\rightarrow$ $\sqcap$ Δ
4 $\rightarrow$ $\diamond$	16 $\rightarrow$ $\sqcap$ $\diamond$
5 $\rightarrow$ $\star$	17 $\rightarrow$ $\sqcap$ $\star$
6 $\rightarrow$ $\wedge$ 0	18 $\rightarrow$ Δ 0
7 $\rightarrow$ $\wedge$ $\wedge$	19 $\rightarrow$ Δ $\wedge$
8 $\rightarrow$ $\wedge$ $\sqcap$	20 $\rightarrow$ Δ $\sqcap$
9 $\rightarrow$ $\wedge$ Δ	21 $\rightarrow$ Δ Δ
10 $\rightarrow$ $\wedge$ $\diamond$	22 $\rightarrow$ Δ $\diamond$
11 $\rightarrow$ $\wedge$ $\star$	23 $\rightarrow$ Δ $\star$

VERTICES $\rightarrow$ DECIMAL

n = number of positions

V = number of vertices at position

$$\text{decimal} = V \times 6^n + \dots + V \times 6^2 + V \times 6^1 + V \times 6^0$$

ex 1: convert $\Delta \star$ to decimal.

$$3 \times 6^1 + 5 \times 6^0$$

$$= 18 + 5$$

$$= 23 \quad \checkmark$$

ex 2: convert $\star \star \diamond \sqcap$ to decimal.

$$5 \times 6^3 + 5 \times 6^2 + 4 \times 6^1 + 2 \times 6^0$$

$$= 1080 + 180 + 24 + 2$$

$$= 1286 \quad \checkmark$$

DECIMAL $\rightarrow$ VERTICES

keep dividing by 6, until you can't anymore. The remainder will be the number of vertices, stop when the solution < 1 .

ex convert 23 to vertices.

$$23 / 6 = 3 \text{ with remainder } 5$$

$$3 / 6 = 3 \text{ with remainder } 3$$

3, 5

$$= \Delta \star \quad \checkmark$$

QUATERNARY SYSTEM

/kwaa·tur·neh·ree/

aka. Base-4, Radix-4

$(0,1,2,3)_4$

Hindu-Arabic Numerals
Originated in India
7th Century AD

More Significant Position
←

#Symbols=4
Radix-4
Base-4

4^0

0

$$0 \times 4^0 = 0$$

1

$$1 \times 4^0 = 1$$

2

$$2 \times 4^0 = 2$$

3

$$3 \times 4^0 = 3$$

More Significant Position
←

#Symbols=4
Radix-4
Base-4

4^1	4^0	
	0	$0 \times 4^0 = 0$
	1	$1 \times 4^0 = 1$
	2	$2 \times 4^0 = 2$
	3	$3 \times 4^0 = 3$
1	0	$1 \times 4^1 + 0 \times 4^0 = 4$
1	1	$1 \times 4^1 + 1 \times 4^0 = 5$
1	2	$1 \times 4^1 + 2 \times 4^0 = 6$
1	3	$1 \times 4^1 + 3 \times 4^0 = 7$
2	0	$2 \times 4^1 + 0 \times 4^0 = 8$

More Significant Position
←

#Symbols=4
Radix-4
Base-4

4^2	4^1	4^0	
	2	0	$2 \times 4^1 + 0 \times 4^0 = 8$
	2	1	$2 \times 4^1 + 1 \times 4^0 = 9$
	2	2	$2 \times 4^1 + 2 \times 4^0 = 10$
	2	3	$2 \times 4^1 + 3 \times 4^0 = 11$
	3	0	$3 \times 4^1 + 0 \times 4^0 = 12$
	3	1	$3 \times 4^1 + 1 \times 4^0 = 13$
	3	2	$3 \times 4^1 + 2 \times 4^0 = 14$
	3	3	$3 \times 4^1 + 3 \times 4^0 = 15$
1	0	0	$1 \times 4^2 + 0 \times 4^1 + 0 \times 4^0 = 16$

More Significant Position
←

#Symbols=4
Radix-4
Base-4

4^3	4^2	4^1	4^0	
		2	0	$2 \times 4^1 + 0 \times 4^0 = 8$
		2	1	$2 \times 4^1 + 1 \times 4^0 = 9$
		2	2	$2 \times 4^1 + 2 \times 4^0 = 10$
		2	3	$2 \times 4^1 + 3 \times 4^0 = 11$
		3	0	$3 \times 4^1 + 0 \times 4^0 = 12$
		3	1	$3 \times 4^1 + 1 \times 4^0 = 13$
		3	2	$3 \times 4^1 + 2 \times 4^0 = 14$
		3	3	$3 \times 4^1 + 3 \times 4^0 = 15$
	1	0	0	$1 \times 4^2 + 0 \times 4^1 + 0 \times 4^0 = 16$

[illegible]

[illegible]

More Significant Position

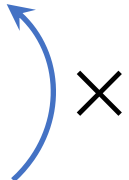


4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0	
3	0	3	0	2	1	3	1	

More Significant Position


4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0	 ×
3	0	3	0	2	1	3	1	
3×4^7	0×4^6	3×4^5	0×4^4	2×4^3	1×4^2	3×4^1	1×4^0	



4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0	
3	0	3	0	2	1	3	1	
3×4^7	0×4^6	3×4^5	0×4^4	2×4^3	1×4^2	3×4^1	1×4^0	Σ
								65,437

[illegible]

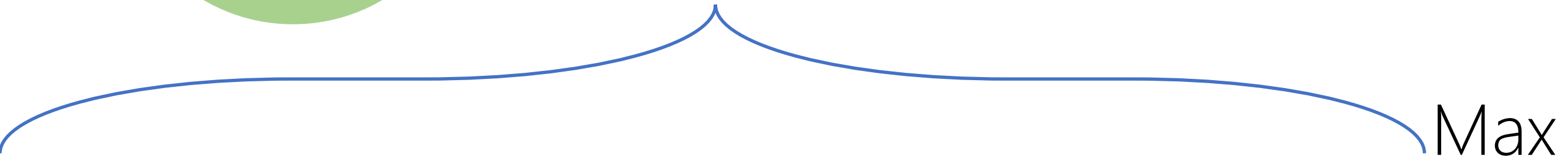
[illegible]

[illegible]

0	0	...				0	0	0
---	---	-----	--	--	--	---	---	---



n positions



3	3	...				3	3	3	?
---	---	-----	--	--	--	---	---	---	---

What is the max number given n position?

$$N = 3 \times 4^{n-1} + 3 \times 4^{n-2} + \dots + 3 \times 4^2 + 3 \times 4^1 + 3 \times 4^0$$

$$N = 3 \times (4^{n-1} + 4^{n-2} + \dots + 4^2 + 4^1 + 4^0) \rightarrow \text{geometric series}$$

$$N = 3 \times \left(\frac{4^n - 1}{4 - 1} \right)$$

$$N = 4^n - 1$$

3n
Hossein's
System

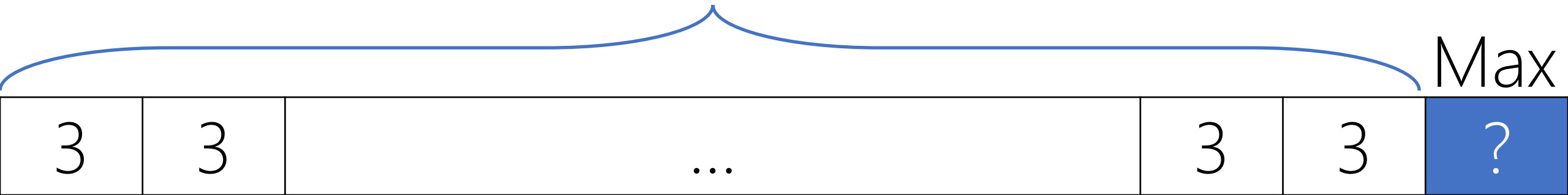
n positions

4^{n-1}	4^{n-2}					4^2	4^1	4^0	
3	3	...				3	3	3	N

What is the max number given n position?

$$n = 10 \Rightarrow 4^{10} - 1 = 1,048,575$$

10 positions



The other way around,

How many positions for a number N ?

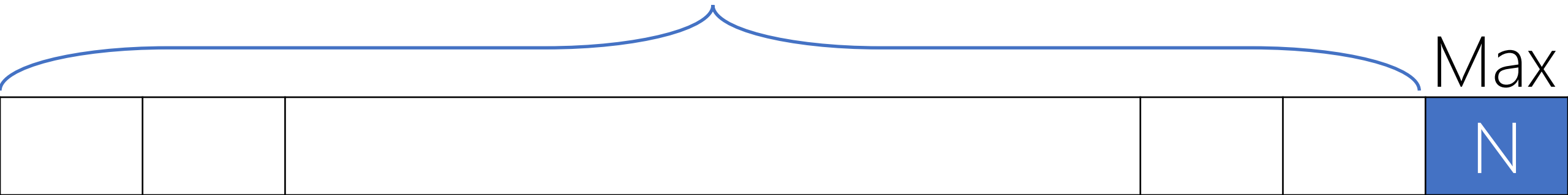
$$4^n - 1 = N$$

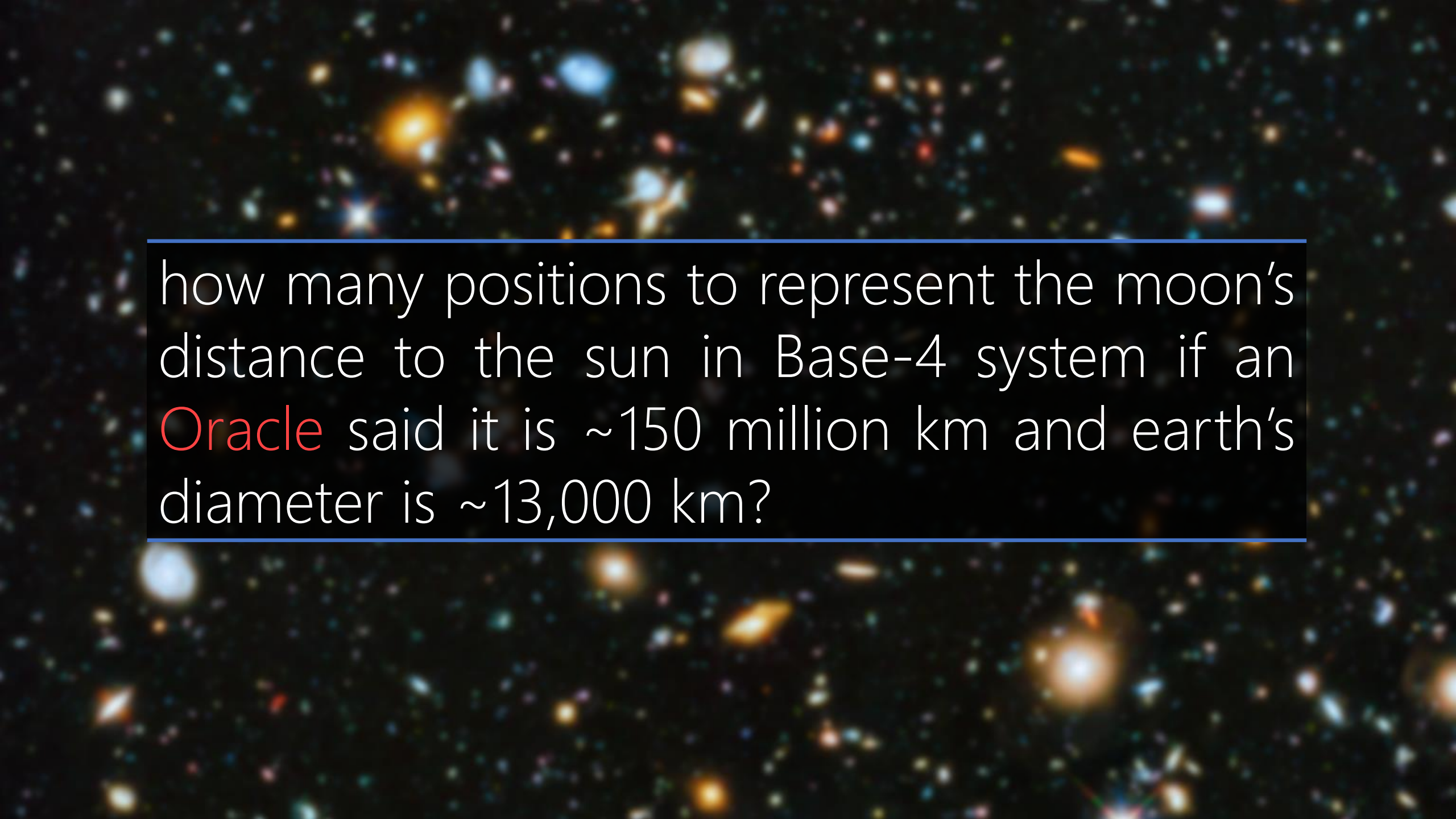
$$4^n = N + 1$$

$$\log_4 4^n = \log(N + 1)$$

$$n = \log_4(N + 1)$$

? positions





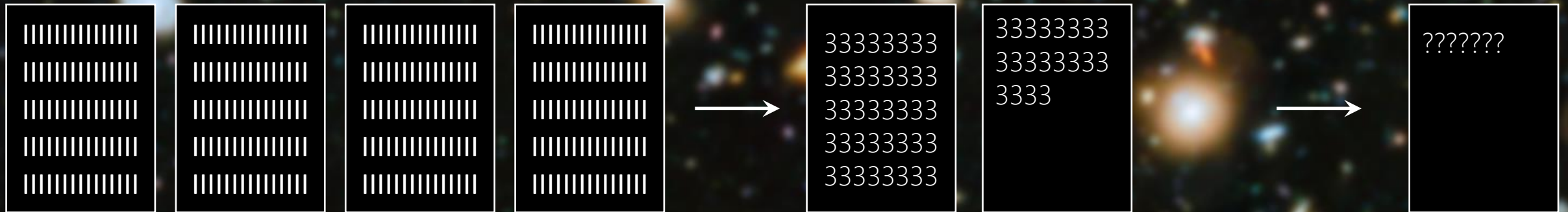
how many positions to represent the moon's distance to the sun in Base-4 system if an **Oracle** said it is ~150 million km and earth's diameter is ~13,000 km?

$\sim 150 \text{ million km} \div \sim 13,000 \text{ km} = \sim 12,000 \text{ Earth}$

$$N = 12,000$$

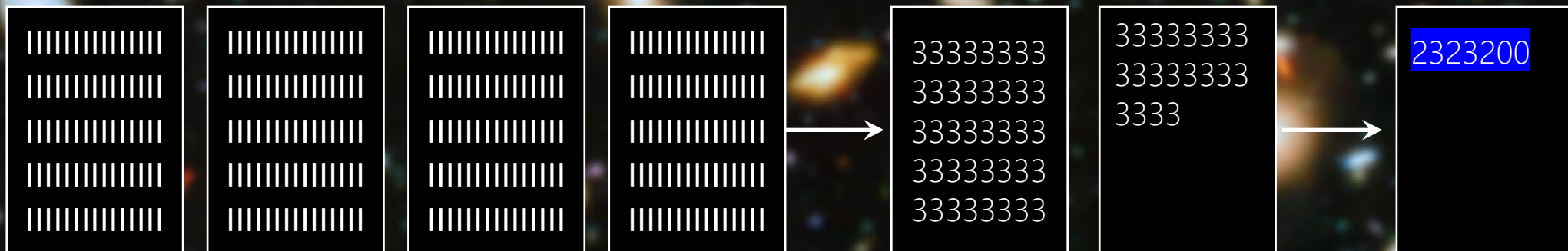
$$n = \text{Log}_4 (12,000+1) = \text{Log}_{10} 12,001 \div \text{Log}_{10} 4 = 4 \div 0.6 = 6.79$$

$\sim 7 \text{ positions}$



$$N = 12,000 \rightarrow (2323200)_4$$

We'll see how to convert from decimal to base-4 or any other number systems later. Stay tuned!



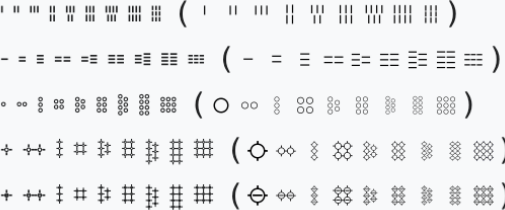
From Wikipedia, the free encyclopedia

For different types of numbers, such as rational numbers, real numbers, complex numbers, etc, see [List of types of numbers](#).

There are many different [numeral systems](#), that is, [writing systems](#) for expressing [numbers](#).

By culture / time period

[\[edit \]](#)

Name	Base	Sample	Approx. First Appearance
	Sort ascending		
Proto-cuneiform numerals	10&60		c. 3500–2000 BCE
Indus numerals			c. 3500–1900 BCE
Proto-Elamite numerals	10&60		3,100 BCE
Sumerian numerals	10&60		3,100 BCE
Egyptian numerals	10		3,000 BCE
Babylonian numerals	10&60		2,000 BCE
Aegean numerals	10		1,500 BCE
Chinese numerals		零一三四五六七八九十百千萬億 (Default, Traditional Chinese)	

Part of a series on

Numeral systems

[Place-value notation](#) [\[show\]](#)

[Sign-value notation](#) [\[show\]](#)

List of numeral systems

[V](#) • [T](#) • [E](#)

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

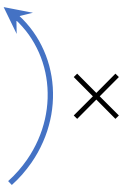
COMMON NUMBER SYSTEMS

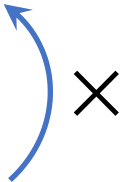
A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines frame the central text.

BINARY | BASE-2 | RADIX-2
 $(0,1)_2$

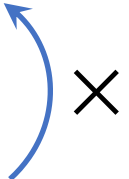
1	0	1	0	1	1	0	1	
---	---	---	---	---	---	---	---	--

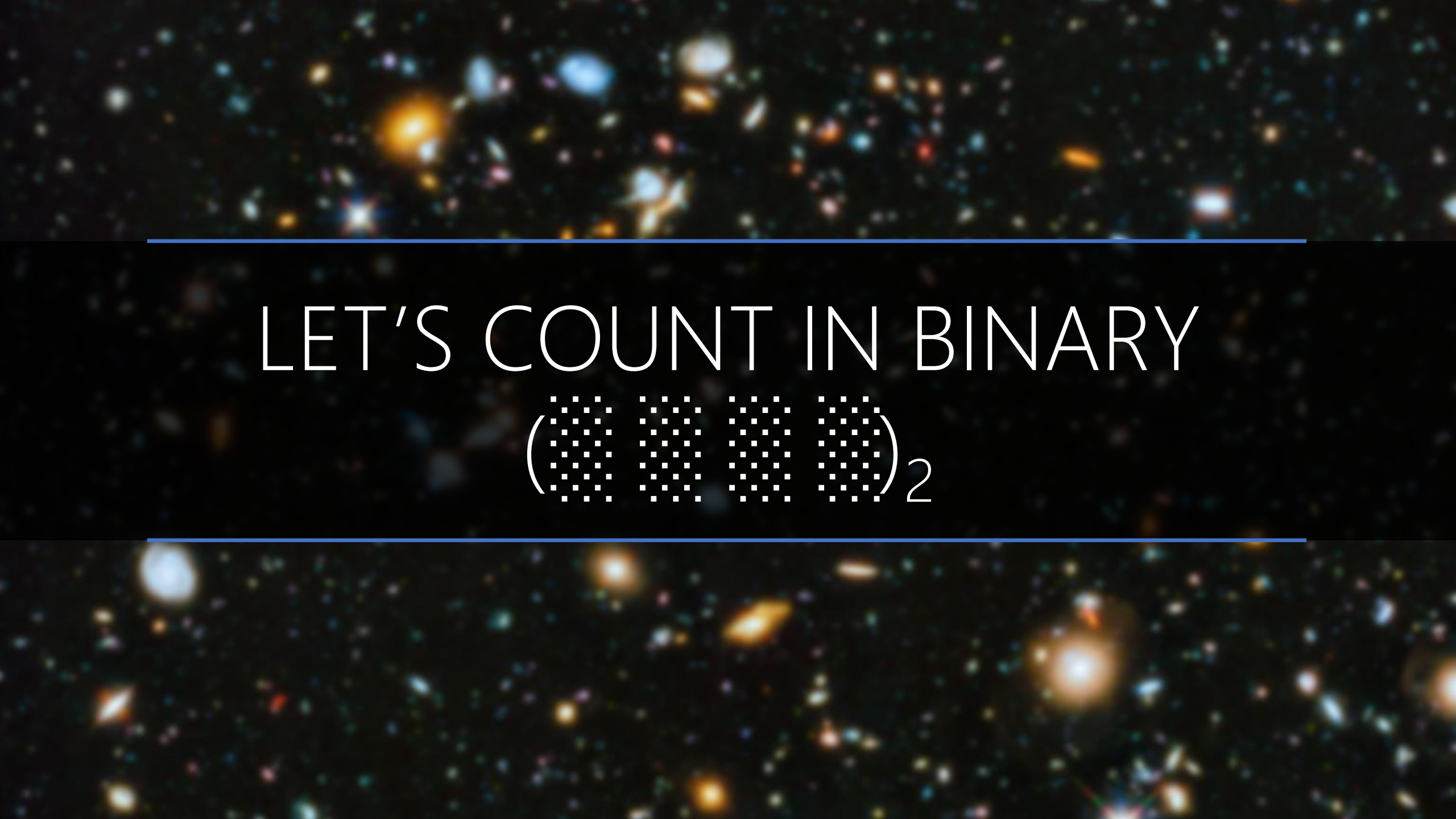
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	Σ
								173

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
3	0	3	0	2	1	3	1	
								Σ

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
3	0	3	0	2	1	3	1	
-	0×2^6	-	0×2^4	-	1×2^2	-	1×2^0	Σ
								Not Valid



LET'S COUNT IN BINARY

$(\begin{array}{c} \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \end{array} \begin{array}{c} \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \end{array} \begin{array}{c} \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \end{array} \begin{array}{c} \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \\ \cdot \cdot \cdot \cdot \cdot \end{array})_2$

LET'S COUNT IN BINARY

$(\begin{array}{|c|c|c|c|} \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \end{array} \begin{array}{|c|c|c|c|} \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \end{array} \begin{array}{|c|c|c|c|} \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \end{array} \begin{array}{|c|c|c|c|} \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \cdot & \cdot & \cdot & \cdot \\ \hline \end{array})_2$

0000, 0001, 0010, 00011, ..., 1101, 1110, 1111

Increment by 1 → Increment by 0001

+ Base-2				1	0	1	1
				0	0	0	1

Increment by 1 → Increment by 0001

						1	
+ Base-2				1	0	1	1
				0	0	0	1
							2→0
							$\frac{2}{2} = 1 \text{ } r \text{ } 0$

Increment by 1 → Increment by 0001

					1	1	
+ Base-2				1	0	1	1
				0	0	0	1
						2→0	0
							$\frac{2}{2} = 1 \text{ } r \text{ } 0$

Increment by 1 → Increment by 0001

					1	1	
+ Base-2				1	0	1	1
				0	0	0	1
				1	1	0	0

https://www.linkedin.com/posts/tom-allen-515058b7_binarysystem-techexplained-digitalworld-activity-7251622787855839232-Rzls/

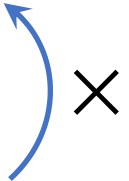


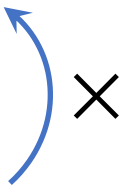
A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. A horizontal blue line runs across the middle of the image, behind the text.

OCTAL | BASE-8 | RADIX-8
 $(0,1,2,3,4,5,6,7)_8$

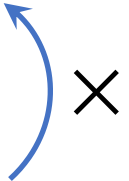
3	0	3	0	2	1	3	1	
---	---	---	---	---	---	---	---	--

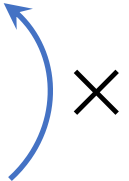
8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	3	0	2	1	3	1	

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	3	0	2	1	3	1	
3×8^7	0×8^6	3×8^5	0×8^4	2×8^3	1×8^2	3×8^1	1×8^0	

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	3	0	2	1	3	1	
3×8^7	0×8^6	3×8^5	0×8^4	2×8^3	1×8^2	3×8^1	1×8^0	Σ
								6,390,873

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	A	0	8	1	3	1	
								Σ

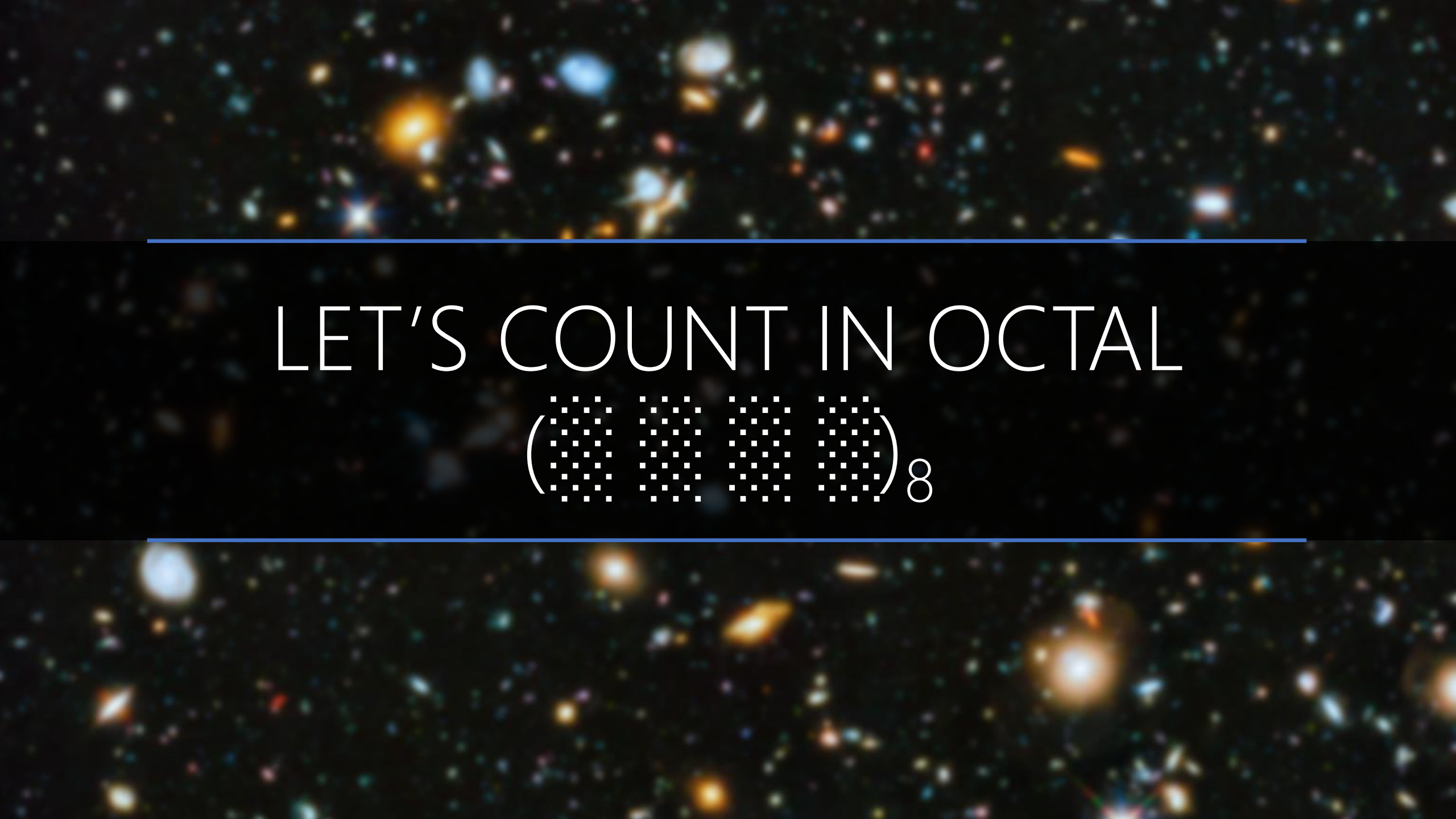
8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
3	0	A	0	8	1	3	1	
3×8^7	0×8^6	-	0×8^4	-	1×8^2	3×8^1	1×8^0	Σ
								Not Valid

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
1	0	1	0	1	1	0	1	
1×8^7	0×8^6	1×8^5	0×8^4	1×8^3	1×8^2	0×8^1	1×8^0	Σ
								2,130,497

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
1	0	1	0	1	1	0	1	
1×8^7	0×8^6	1×8^5	0×8^4	1×8^3	1×8^2	0×8^1	1×8^0	
								2,130,497

Radix-8
vs.
Radix-2

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	
								173



LET'S COUNT IN OCTAL

$(\begin{smallmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{smallmatrix} \begin{smallmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{smallmatrix} \begin{smallmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{smallmatrix} \begin{smallmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{smallmatrix})_8$



LET'S COUNT IN OCTAL

$(\begin{smallmatrix} \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet \end{smallmatrix})_8$

0000, 0001, 0002, ..., 0007, 0010, ..., 7775, 7776, 7777

Increment by 1 → Increment by 0001

+ Base-8				5	2	7	7
				0	0	0	1

Increment by 1 $\rightarrow$ Increment by 0001

						1	
+ Base-8				5	2	7	7
				0	0	0	1
							8→0

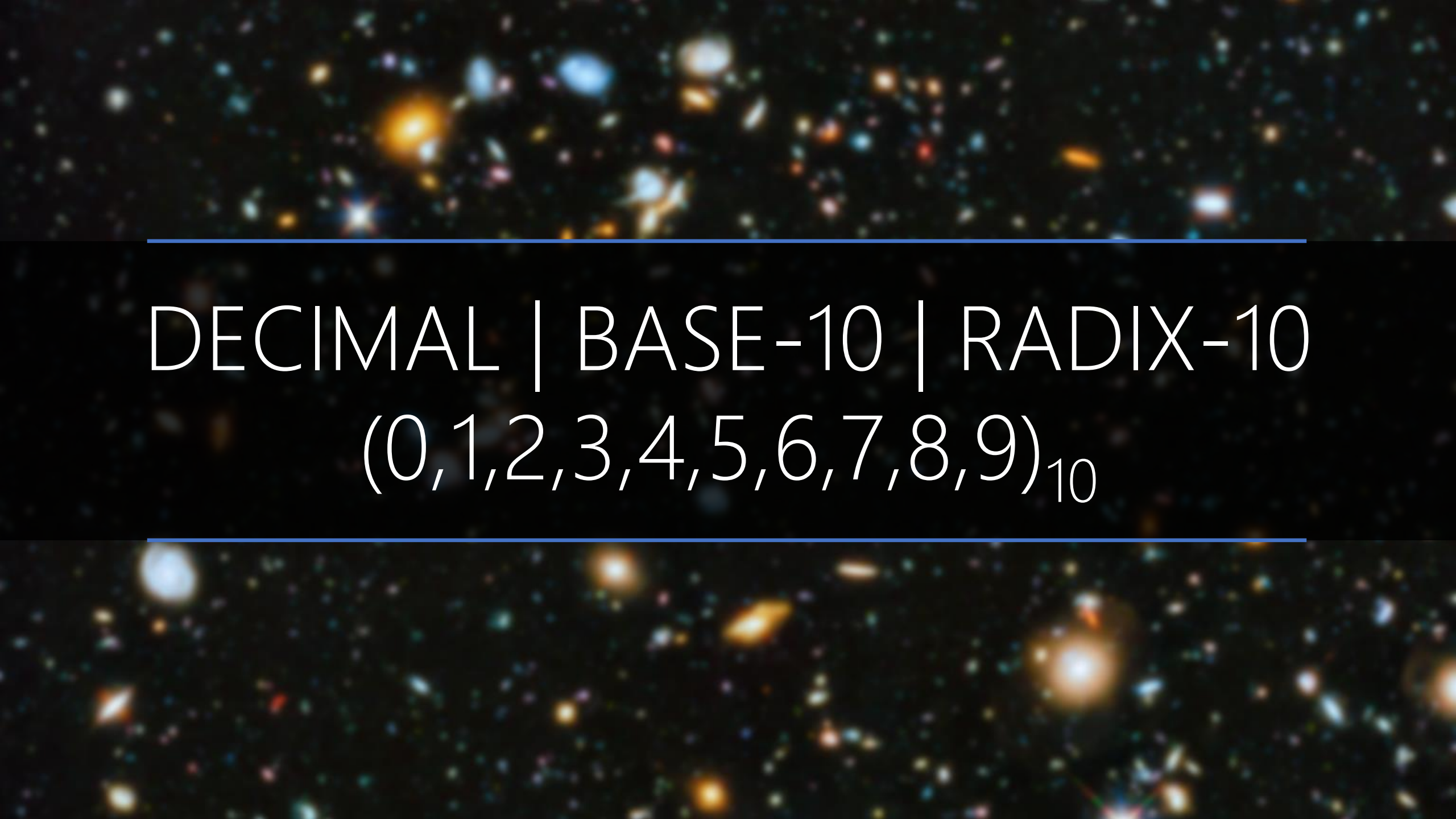
$$\frac{8}{8} = 1 \text{ r } 0$$

Increment by 1 → Increment by 0001

					1	1	
+ Base-8				5	2	7	7
				0	0	0	1
						8→0	0
						$\frac{8}{8} = 1 \text{ } r \text{ } 0$	

Increment by 1 → Increment by 0001

					1	1	
+ Base-8				5	2	7	7
				0	0	0	1
				5	3	0	0

A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. A solid black horizontal band runs across the middle of the image, serving as a backdrop for the text.

DECIMAL | BASE-10 | RADIX-10
 $(0,1,2,3,4,5,6,7,8,9)_{10}$

Arabic

0123456789

Eastern Arabic

٠١٢٣٤٥٦٧٨٩

Roman

I II III IV V VI VII VIII IX X

Bengali–Assamese

Why?

০ ১ ২ ৩ ৪ ৫ ৬ ৭ ৮ ৯

Malayalam

൦ ൧ ൨ ൩ ൪ ൫ ൬ ൭ ൮ ൯

Thai

๐ ๑ ๒ ๓ ๔ ๕ ๖ ๗ ๘ ๙

Chinese

〇 一 二 三 四 五 六 七 八 九

Arabic

Eastern Arabic

0123456789

٠ ١ ٢ ٣ ٤ ٥ ٦ ٧ ٨ ٩

*"the **hand** makes the two complementary aspects of integers entirely intuitive. It serves as an instrument permitting natural movement between cardinal and ordinal numbering. If you need to show that a set contains three, four, seven or ten elements, you raise or bend **simultaneously** three, four, seven or ten fingers, using your hand as cardinal mapping. If you want to count out the same things, then you bend or raise three, four, seven or ten fingers in **succession**, using the hand as an ordinal counting tool."*

- Georges Ifrah The Universal History of Numbers (Wiley, 2000, pp. 21-22)

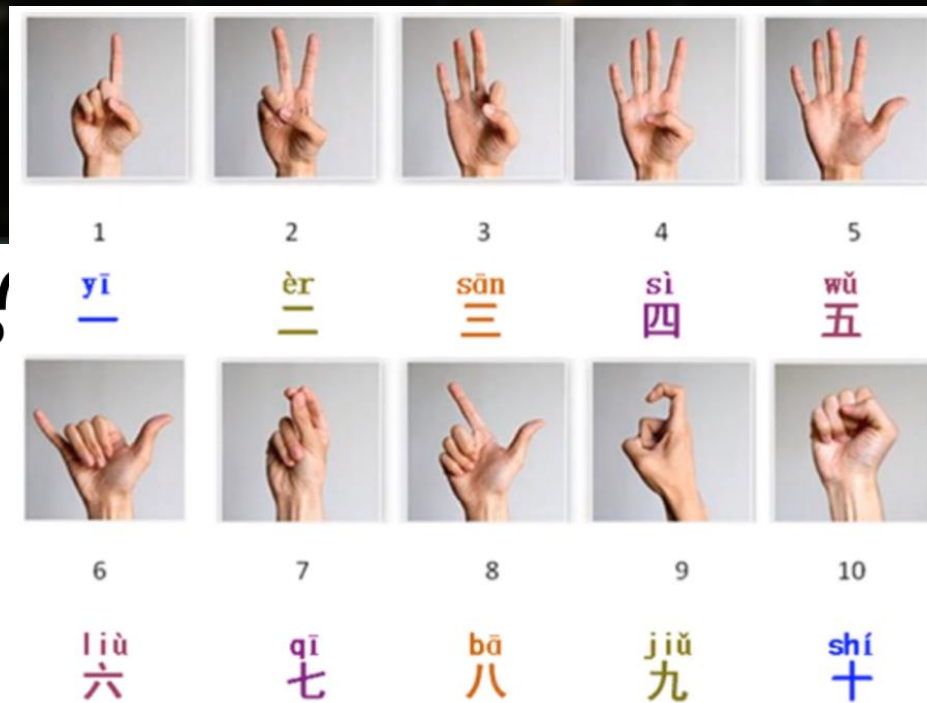
Thanks Cecelia Nydam and Giavi Tran for the hint!

Thai

Chinese

๐ ๑ ๒ ๓ ๔

〇 一 二 三



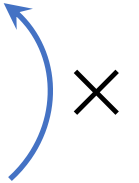
3	0	3	0	2	1	3	1	
---	---	---	---	---	---	---	---	--

10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
3	0	3	0	2	1	3	1	

10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
3	0	3	0	2	1	3	1	
3×10^7	0×10^6	3×10^5	0×10^4	2×10^3	1×10^2	3×10^1	1×10^0	

10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
3	0	3	0	2	1	3	1	
3×10^7	0×10^6	3×10^5	0×10^4	2×10^3	1×10^2	3×10^1	1×10^0	Σ
								30,302,131

10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
3	0	3	0	2	1	3	1	
3×10^7	0×10^6	3×10^5	0×10^4	2×10^3	1×10^2	3×10^1	1×10^0	Σ
								30,302,131

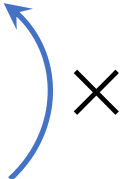
10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
1	0	1	0	1	1	0	1	
1×10^7	0×10^6	1×10^5	0×10^4	1×10^3	1×10^2	0×10^1	1×10^0	Σ
								10,101,101

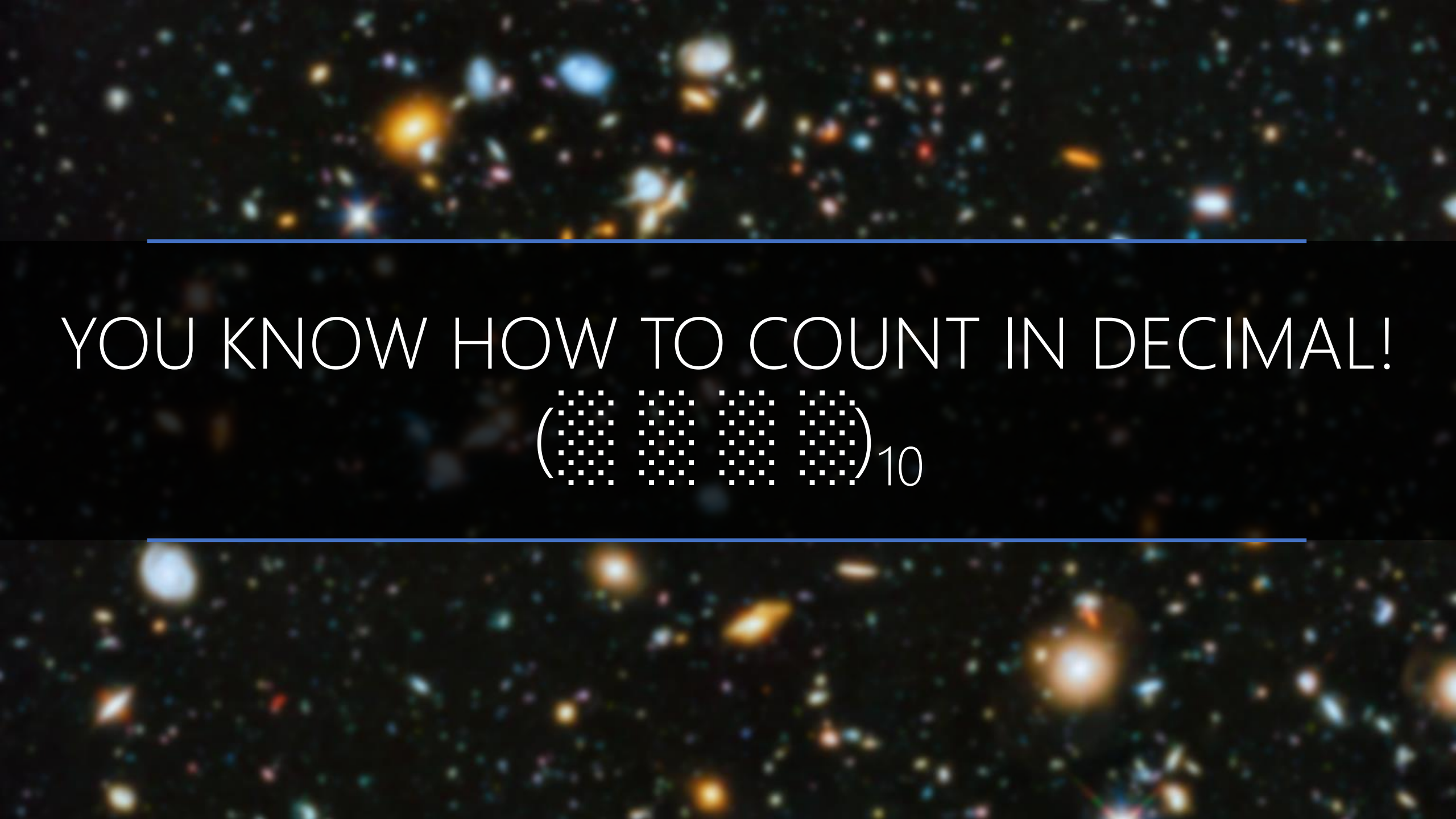
10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
1	0	1	0	1	1	0	1	
1×10^7	0×10^6	1×10^5	0×10^4	1×10^3	1×10^2	0×10^1	1×10^0	
								10,101,101

8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
1	0	1	0	1	1	0	1	
1×8^7	0×8^6	1×8^5	0×8^4	1×8^3	1×8^2	0×8^1	1×8^0	
								2,130,497

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	
								173

10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
3	0	A	0	8	1	3	1	$\times$
								Σ

10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
3	0	A	0	8	1	3	1	
3×10^7	0×10^6	-	0×10^4	8×10^3	1×10^2	3×10^1	1×10^0	Σ
								Not Valid



YOU KNOW HOW TO COUNT IN DECIMAL!

$$(\begin{array}{c} \cdot \cdot \cdot \\ \cdot \cdot \cdot \\ \cdot \cdot \cdot \end{array} \begin{array}{c} \cdot \cdot \cdot \\ \cdot \cdot \cdot \\ \cdot \cdot \cdot \end{array} \begin{array}{c} \cdot \cdot \cdot \\ \cdot \cdot \cdot \\ \cdot \cdot \cdot \end{array} \begin{array}{c} \cdot \cdot \cdot \\ \cdot \cdot \cdot \\ \cdot \cdot \cdot \end{array})_{10}$$



We Should All Be Counting in Base-12

Jovan Djokic posted Sep 19, 2023 5:33 PM • 242 Words [★ Subscribed](#)



The Base-12 (a.k.a. "duodecimal", or "dozenal") number system is superior in so many ways to the base 10 number system that we use in day-to-day life. It is no harder to use to count once you have gotten used to it, and in fact it makes many math operations radically easier and cleaner. For example, 12 is cleanly divisible (no remainder) by 2, 3, 4, 6, and 12, whereas 10 is only divisible by 2, 5, and 10. This makes simple calculations (especially mental math) so much easier as more often than not any sort of division will result in an easily reducible fraction rather than a jumbled mess of decimals. It is hard to fathom that fractions as simple and prevalent as $\frac{1}{3}$ (one third) or $\frac{1}{4}$ (one quarter) can not be easily represented by our numbering system. Base-12 solves all of this and makes math so much more intuitive.

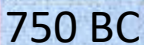
Finally, a common complaint about switching to base-12 is that it would be harder to teach to children since they normally use their fingers to count, but this is easily solved by using the knuckles of 4 fingers to count (3 knuckles per finger X 4 non-thumb fingers per hand). To insist that base-10 is a superior number system to base-12 is nothing but an appeal to tradition and an unwillingness to learn outside of one's comfort zone.

We should all be counting in base-12, though sadly we likely never will.



0	1	2	3	4
	•	••	•••	••••
5	6	7	8	9
	•	••	•••	••••
10	11	12	13	14
	•	••	•••	••••
15	16	17	18	19
	•	••	•••	••••

Maya cities developed ~750 BC, and by 500 BC these cities possessed monumental architecture. A vigesimal (/vɪˈdʒɛsɪməl/) or base-20 (base-score) numeral system



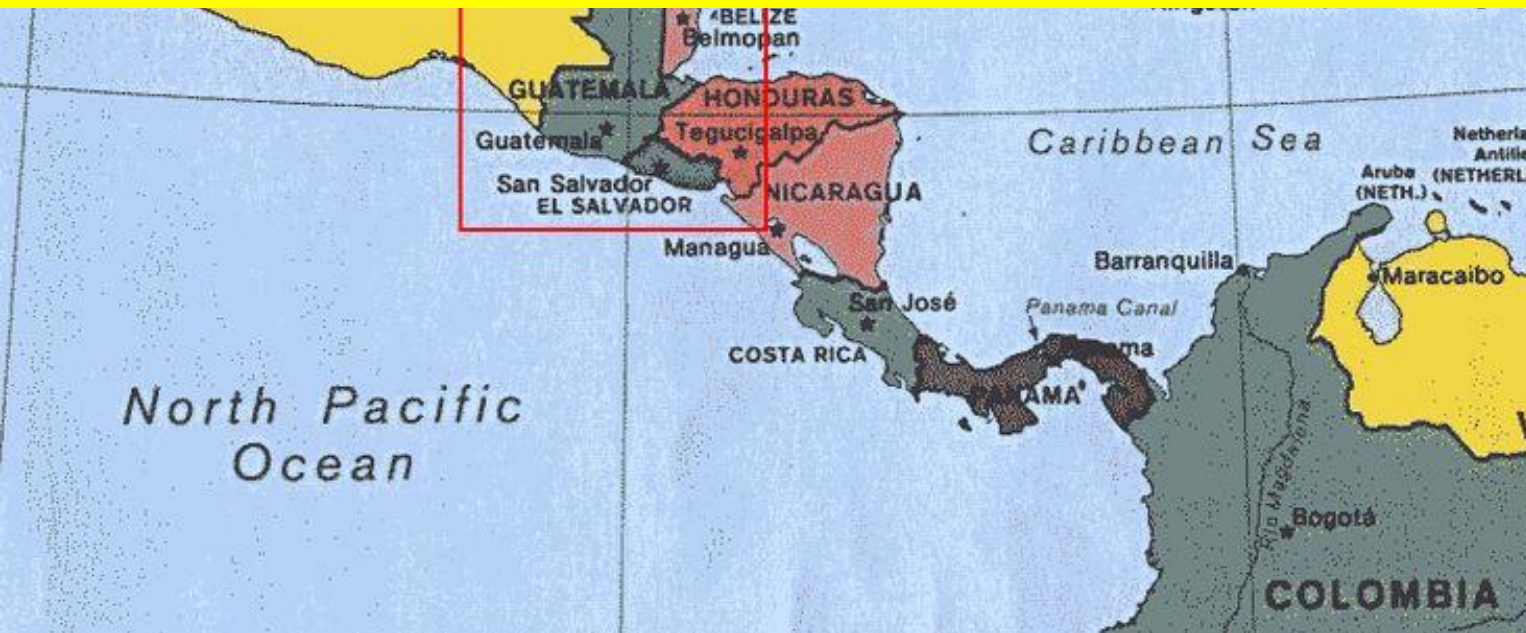
400s			
20s	●	●	● =====
1s	●●● =====	●●●● =====	=====
	33	429	?

https://en.wikipedia.org/wiki/Maya_numerals



0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

BUT WHY?

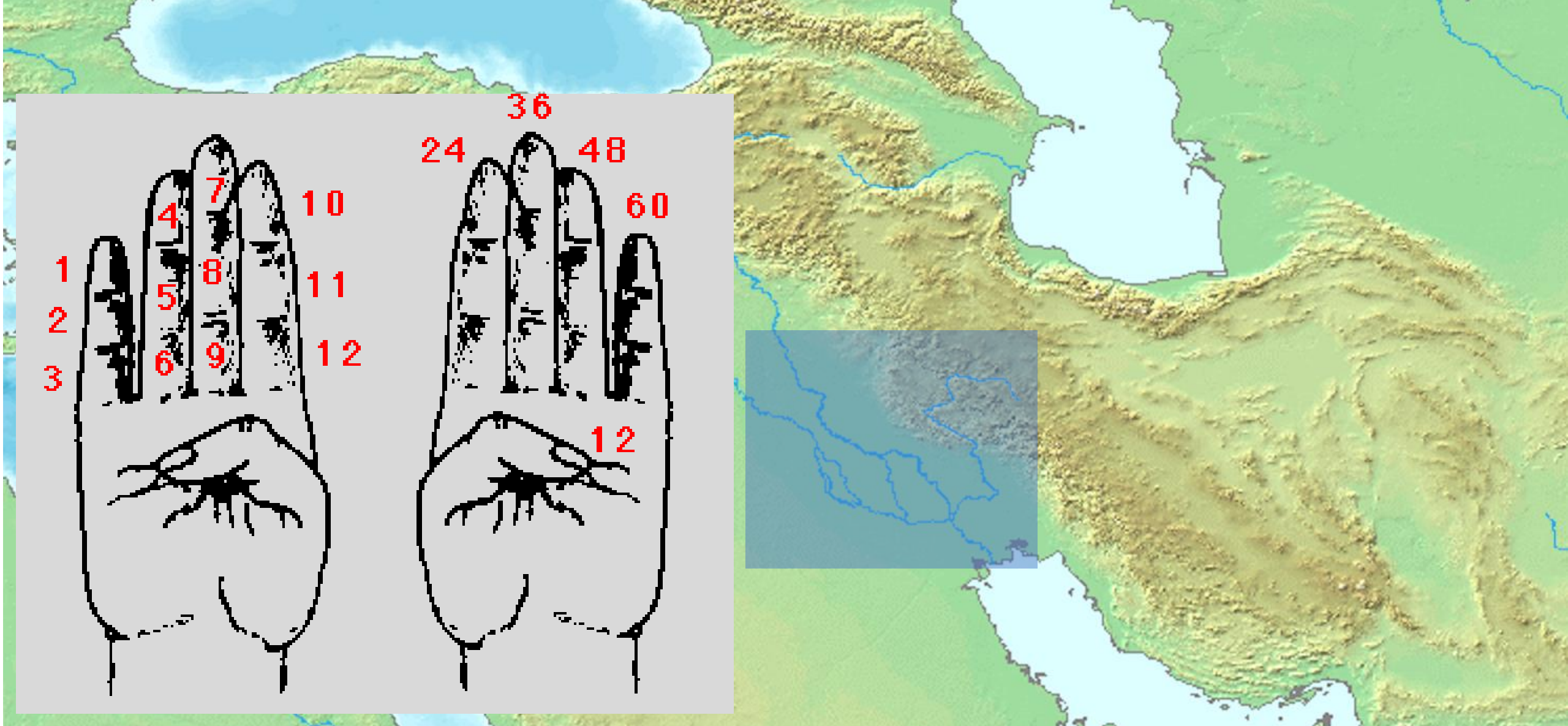


400s			
20s			
1s			
	33	429	?

https://en.wikipedia.org/wiki/Maya_numerals

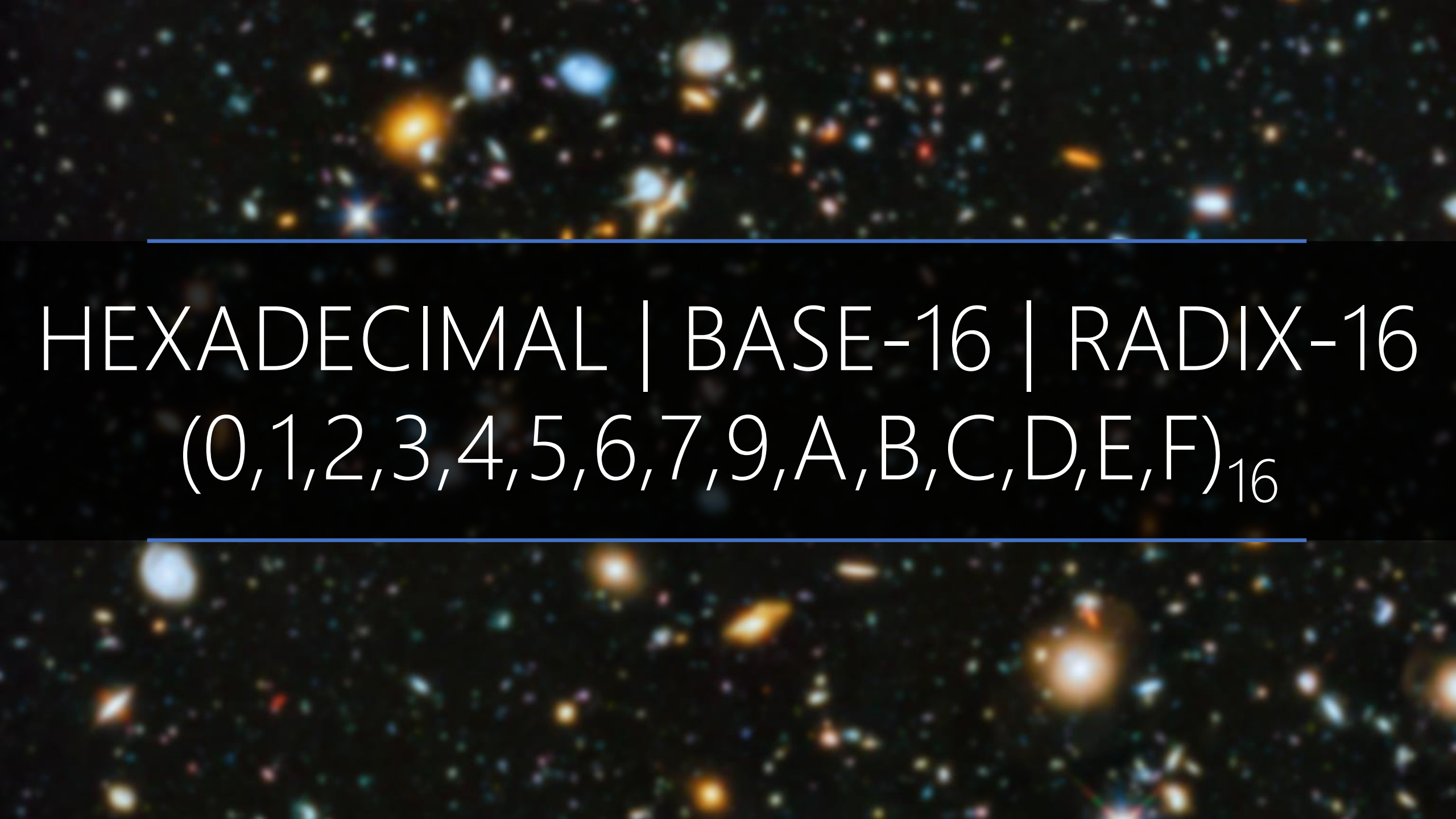


$$\begin{array}{ccccc} \text{—} & + & \text{●●●} & = & \text{●●●} \\ & & \text{—} & & \text{—} \\ \text{●●●} & - & \text{—} & = & \text{●●●} \\ \text{—} & & & & \text{—} \end{array}$$

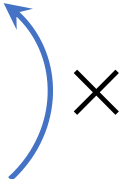


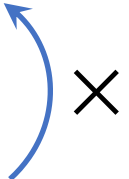
Sumer (/ˈsu:mər/), one of the earliest known civilization (south-central Iraq), 6-5 millennium BC. Sexagesimal, Base-60 or Sexagenary (<https://en.wikipedia.org/wiki/Sexagesimal>)

Weird NUMBER SYSTEMS
Common in CS



HEXADECIMAL | BASE-16 | RADIX-16
(0,1,2,3,4,5,6,7,9,A,B,C,D,E,F)₁₆

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
3	0	A	0	9	1	3	1	
								Σ

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
3	0	A	0	9	1	3	1	
3×16^7	0×16^6	$A \times 16^5$	0×16^4	9×16^3	1×16^2	3×16^1	1×16^0	Σ

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
3	0	A	0	9	1	3	1	
3×16^7	0×16^6	$A \times 16^5$	0×16^4	9×16^3	1×16^2	3×16^1	1×16^0	Σ

 $A = (9 + 1) = (10)_{10}$

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	 ×
3	0	A	0	9	1	3	1	
3×16^7	0×16^6	$A \times 16^5$	0×16^4	9×16^3	1×16^2	3×16^1	1×16^0	Σ
								815,829,297

 $A = (9 + 1) = (10)_{10}$

$$1,2,3,4,5,6,7,8,9,A = 9 + 1 = (10)_{10}$$

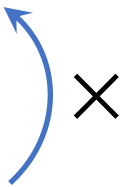
$$B = A + 1 = (11)_{10}$$

$$C = B + 1 = (12)_{10}$$

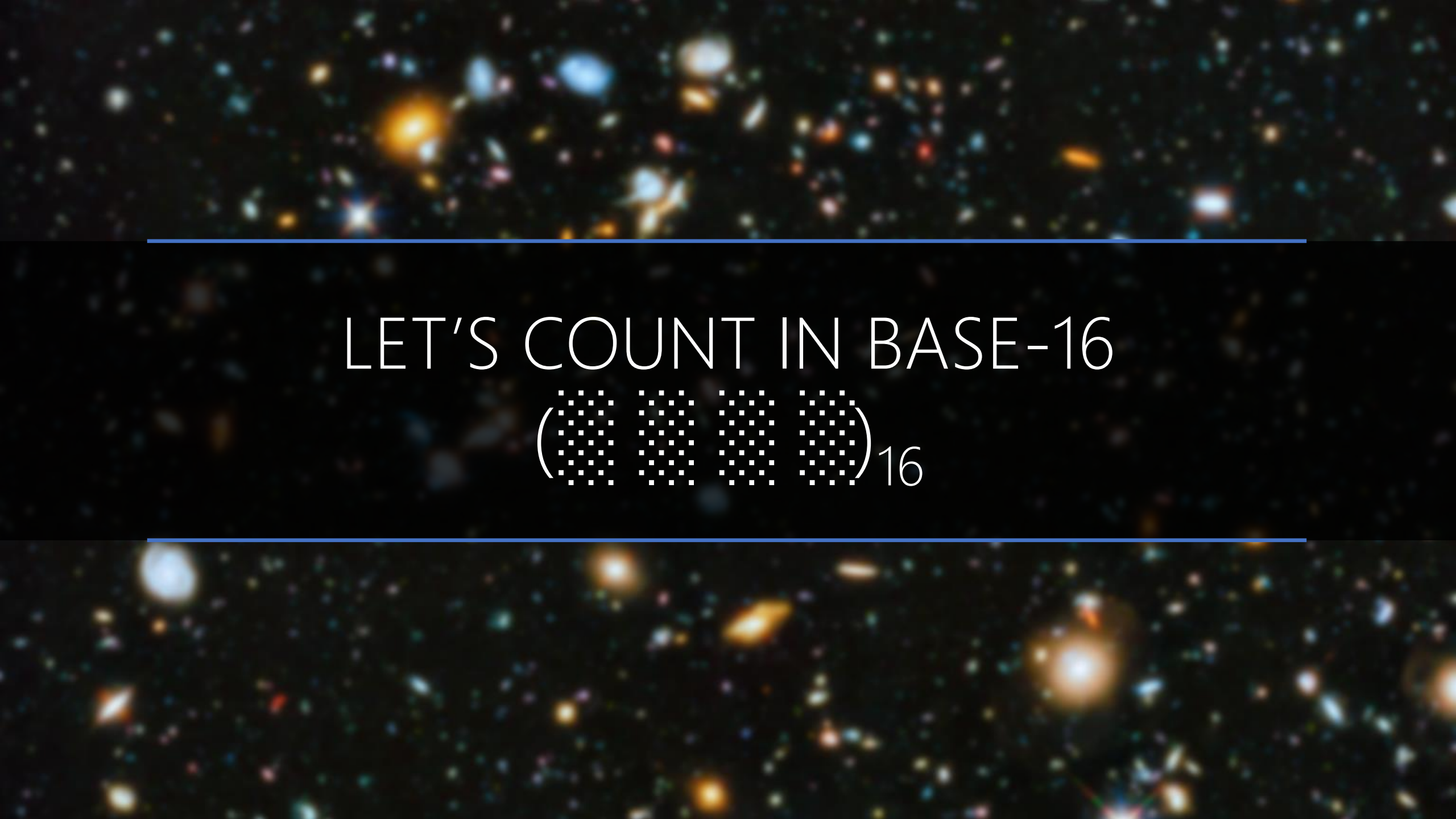
$$D = C + 1 = (13)_{10}$$

$$E = D + 1 = (14)_{10}$$

$$F = E + 1 = (15)_{10}$$

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
1	0	1	0	1	1	0	1	
1×16^7	0×16^6	1×16^5	0×16^4	1×16^3	1×16^2	0×16^1	1×16^0	Σ
								269,488,385

16^7	16^6	16^5	16^4	16^3	16^2	16^1	16^0	
1	0	1	0	1	1	0	1	
1×16^7	0×16^6	1×16^5	0×16^4	1×16^3	1×16^2	0×16^1	1×16^0	
								269,488,385
10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0	
1	0	1	0	1	1	0	1	
1×10^7	0×10^6	1×10^5	0×10^4	1×10^3	1×10^2	0×10^1	1×10^0	
								10,101,101
8^7	8^6	8^5	8^4	8^3	8^2	8^1	8^0	
1	0	1	0	1	1	0	1	
1×8^7	0×8^6	1×8^5	0×8^4	1×8^3	1×8^2	0×8^1	1×8^0	
								2,130,497
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
1	0	1	0	1	1	0	1	
1×2^7	0×2^6	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	
								173

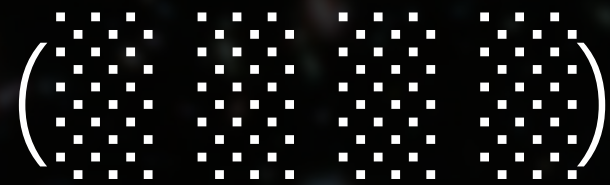


LET'S COUNT IN BASE-16

(   )₁₆



LET'S COUNT IN BASE-16

()₁₆

0000, 0001, 0002, ..., 000F, 0010, ..., FFFD, FFFE, FFFF

Increment by 1 → Increment by 0001

+ Base-16				5	C	A	F
				0	0	0	1

Increment by 1 → Increment by 0001

						1	
+ Base-16				5	C	A	F ₌₁₅
				0	0	0	1
							16→0

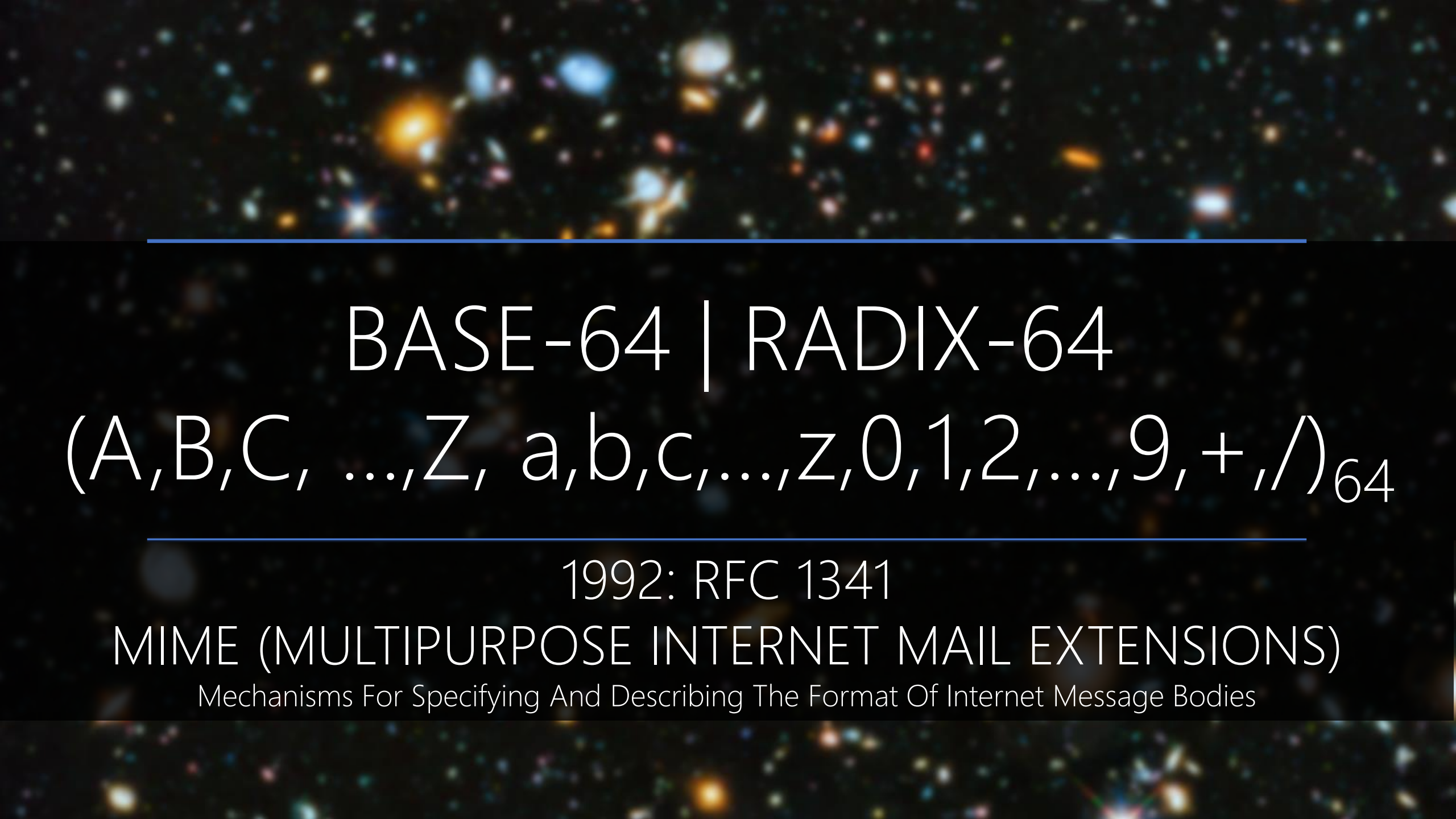
$\frac{16}{16} = 1 \text{ } r \text{ } 0$

Increment by 1 → Increment by 0001

+ Base-16				5	C	1 A ₌₁₀	F ₌₁₅
				0	0	0	1
						11→B	0

Increment by 1 → Increment by 0001

+ Base-16				5	C	A ₌₁₀	F ₌₁₅
				0	0	0	1
				5	C	B	0



BASE-64 | RADIX-64
(A,B,C, ...,Z, a,b,c,...,z,0,1,2,...,9,+ ,/)₆₄

1992: RFC 1341

MIME (MULTIPURPOSE INTERNET MAIL EXTENSIONS)

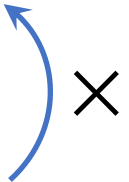
Mechanisms For Specifying And Describing The Format Of Internet Message Bodies

Digit	Value		Digit	Value		Digit	Value		Digit	Value
A	0		Q	16		g	32		w	48
B	1		R	17		h	33		x	49
C	2		S	18		i	34		y	50
D	3		T	19		j	35		z	51
E	4		U	20		k	36		0	52
F	5		V	21		l	37		1	53
G	6		W	22		m	38		2	54
H	7	→	X	23	→	n	39	→	3	55
I	8		Y	24		o	40		4	56
J	9		Z	25		p	41		5	57
K	10		a	26		q	42		6	58
L	11		b	27		r	43		7	59
M	12		c	28		s	44		8	60
N	13		d	29		t	45		9	61
O	14		e	30		u	46		+	62
P	15		f	31		v	47		/	63

3	a	A	/	d	1	H	+	
---	---	---	---	---	---	---	---	--

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
								Σ

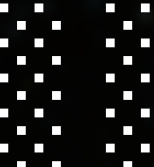
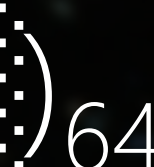
Digit	Value		Digit	Value		Digit	Value		Digit	Value
A	0	→	Q	16	→	g	32	→	w	48
B	1		R	17		h	33		x	49
C	2		S	18		i	34		y	50
D	3		T	19		j	35		z	51
E	4		U	20		k	36		0	52
F	5		V	21		l	37		1	53
G	6		W	22		m	38		2	54
H	7		X	23		n	39		3	55
I	8		Y	24		o	40		4	56
J	9		Z	25		p	41		5	57
K	10		a	26		q	42		6	58
L	11		b	27		r	43		7	59
M	12		c	28		s	44		8	60
N	13		d	29		t	45		9	61
O	14		e	30		u	46		+	62
P	15		f	31		v	47		/	63

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
55 $\times 64^7$	26 $\times 64^6$	0 $\times 64^5$	63 $\times 64^4$	29 $\times 64^3$	53 $\times 64^2$	7 $\times 64^1$	62 $\times 64^0$	Σ

64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	
55 $\times 64^7$	26 $\times 64^6$	0 $\times 64^5$	63 $\times 64^4$	29 $\times 64^3$	53 $\times 64^2$	7 $\times 64^1$	62 $\times 64^0$	Σ
								243,680,329, 290,238

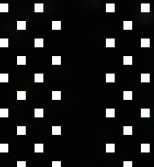
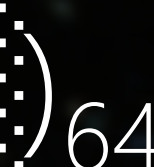


LET'S COUNT IN BASE-64

(   )₆₄



LET'S COUNT IN BASE-64

(   )₆₄

AAAA, AAAB, AAAC, ..., AAA/, AABA, ..., ///9, ///+, ////

Increment by 1 → Increment by AAAB

+ Base-64				a	+	Z	/
				A=0	A=0	A=0	B=1

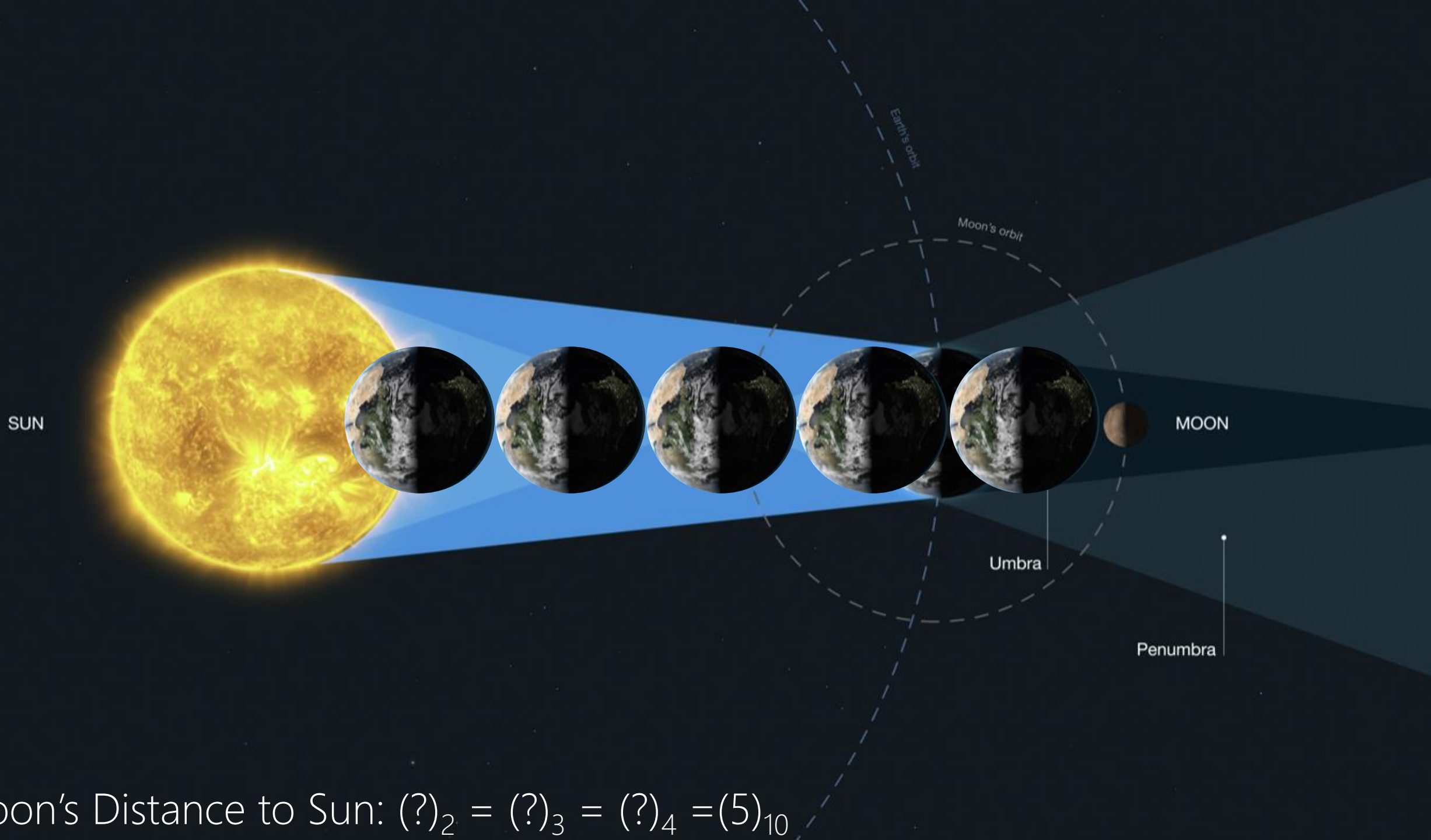
Increment by 1 → Increment by **AAAB**

+ Base-64				a ₌₂₆	+ ₌₆₂	Z ₌₂₅	/ ₌₆₃
				A ₌₀	A ₌₀	A ₌₀	B ₌₁

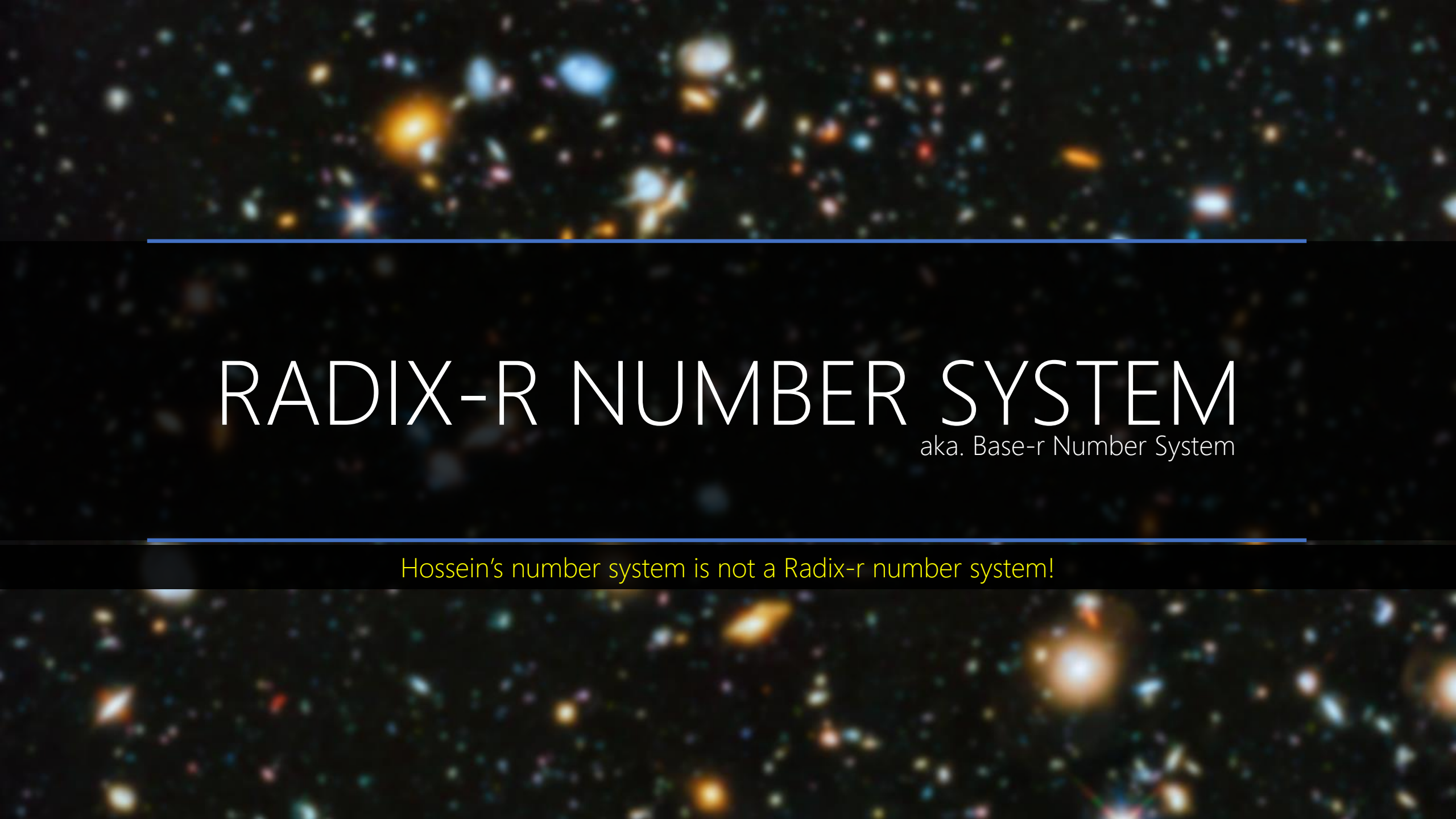
+ Base-64				a ₌₂₆	+ ₌₆₂	B ₌₁	
				A ₌₀	A ₌₀	Z ₌₂₅	/ ₌₆₃
						A ₌₀	B ₌₁
							A ₌₀

$$\frac{64}{64} = 1 \text{ r } 0$$

+ Base-64				a ₌₂₆	+ ₌₆₂	B ₌₁	
				A ₌₀	A ₌₀	Z ₌₂₅	/ ₌₆₃
				a	+	a ₌₂₆	A ₌₀



Moon's Distance to Sun: $(?)_2 = (?)_3 = (?)_4 = (5)_{10}$



RADIX-R NUMBER SYSTEM

aka. Base-r Number System

Hosseini's number system is not a Radix-r number system!

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1} r^{n-1} + d_{n-2} r^{n-2} + \dots + d_i r^i + \dots + d_2 r^2 + d_1 r^1 + d_0 r^0)_{10}$$

where:

r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight (significance) of position i

n = number of digits in N

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where:

r = radix (base)

d_i = digit at position i , $0 \leq d_i \leq r - 1$

r^i = weight (significance) of position i

n = number of digits in N

Let $(N)_r$ be a radix- r (base- r) number in a positional weighting number system, then

Min	$= (0_{n-1}0_{n-2} \cdots 0_10_0)_r$	$= (0)_{10}$
Max	$= ((r-1)_{n-1}(r-1)_{n-2} \cdots (r-1)_1(r-1)_0)_r$	$= (r^n - 1)_{10}$
Unit	$= (0_{n-1}0_{n-2} \cdots 0_11_0)_r$	$= (1)_{10}$

where:

- r = radix (base)
- r^i = weight of position i
- n = number of digits in N



FRACTION

